

Printer Dover

Spruce version 11.2 -- spooler version 11.2

File: Procl-apcRev-Da.sil etc.

Creation date: 8-Oct-81 17:45:06

For: Spruce

34 total sheets = 33 pages, 1 copy.

Problems encountered:

Font HELVETICA24B substituted for font HELVE24.

Character code 117b not found in font Gates32.

Character code 157b not found in font Gates32.

Character code 117b not found in font Gates32.

Character code 157b not found in font Gates32.

Character code 117b not found in font Gates32.

Character code 157b not found in font Gates32.

Character code 117b not found in font Gates32.

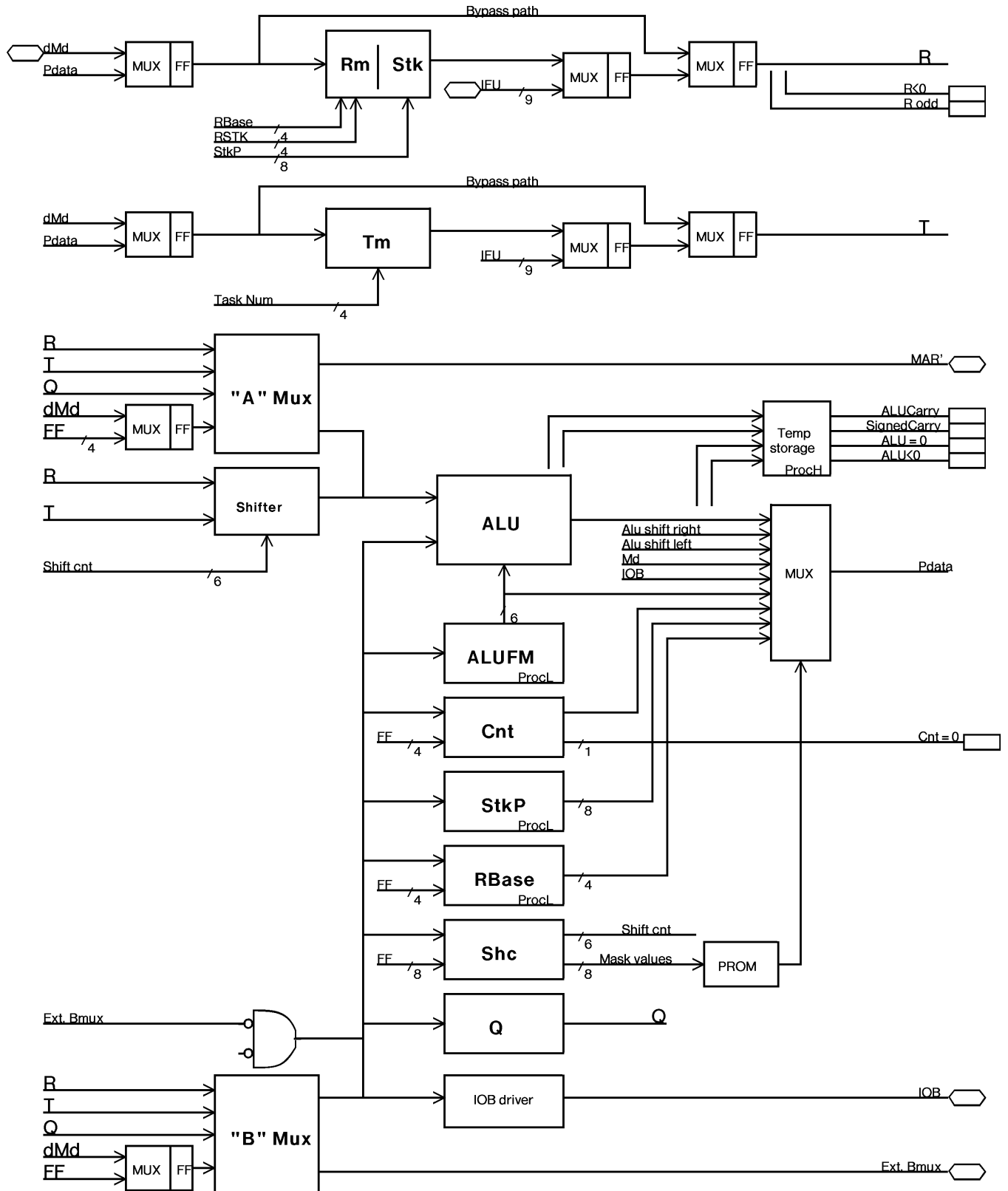
DORADO SCHEMATICS

Low Byte

PROCESSOR

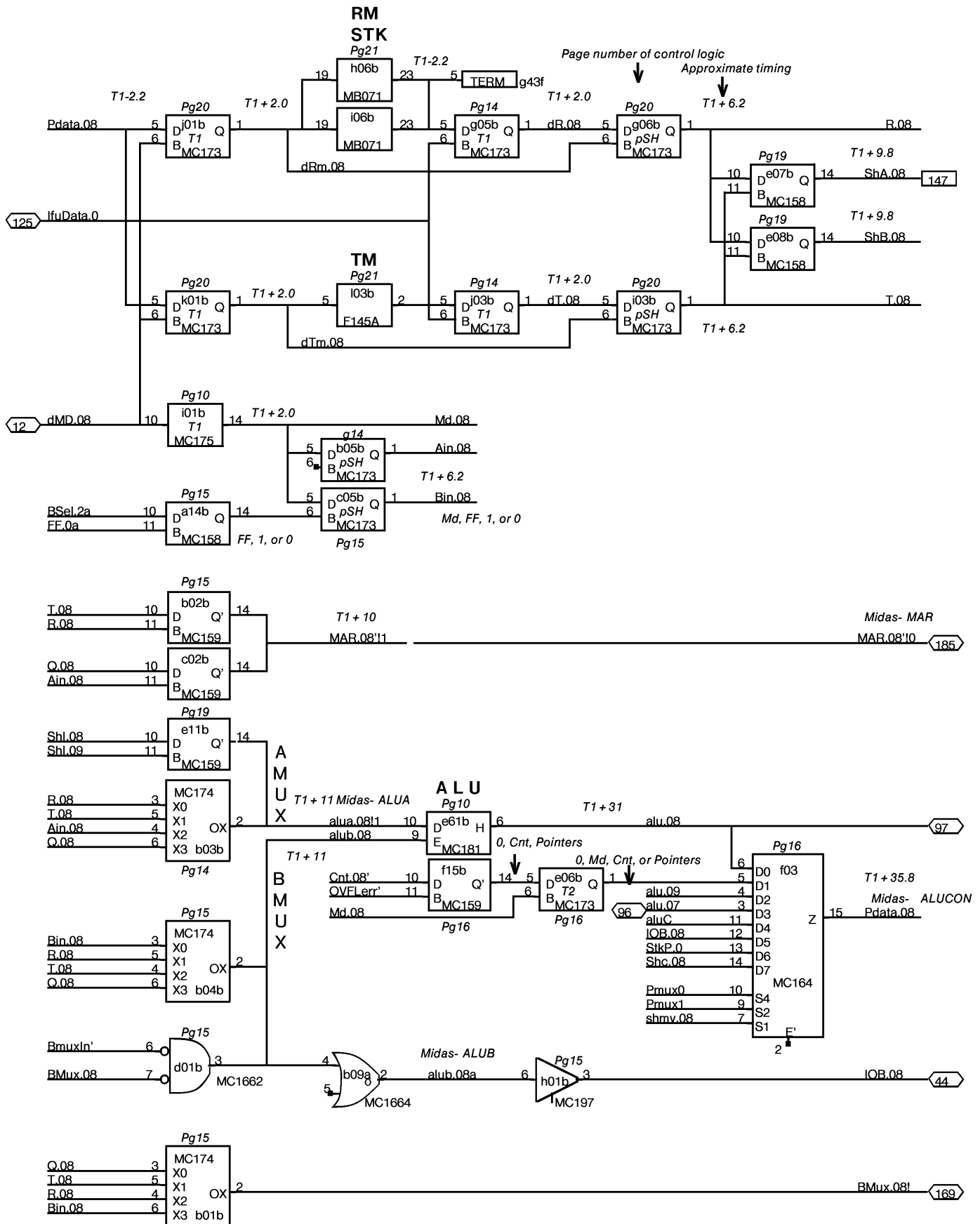
Table of contents

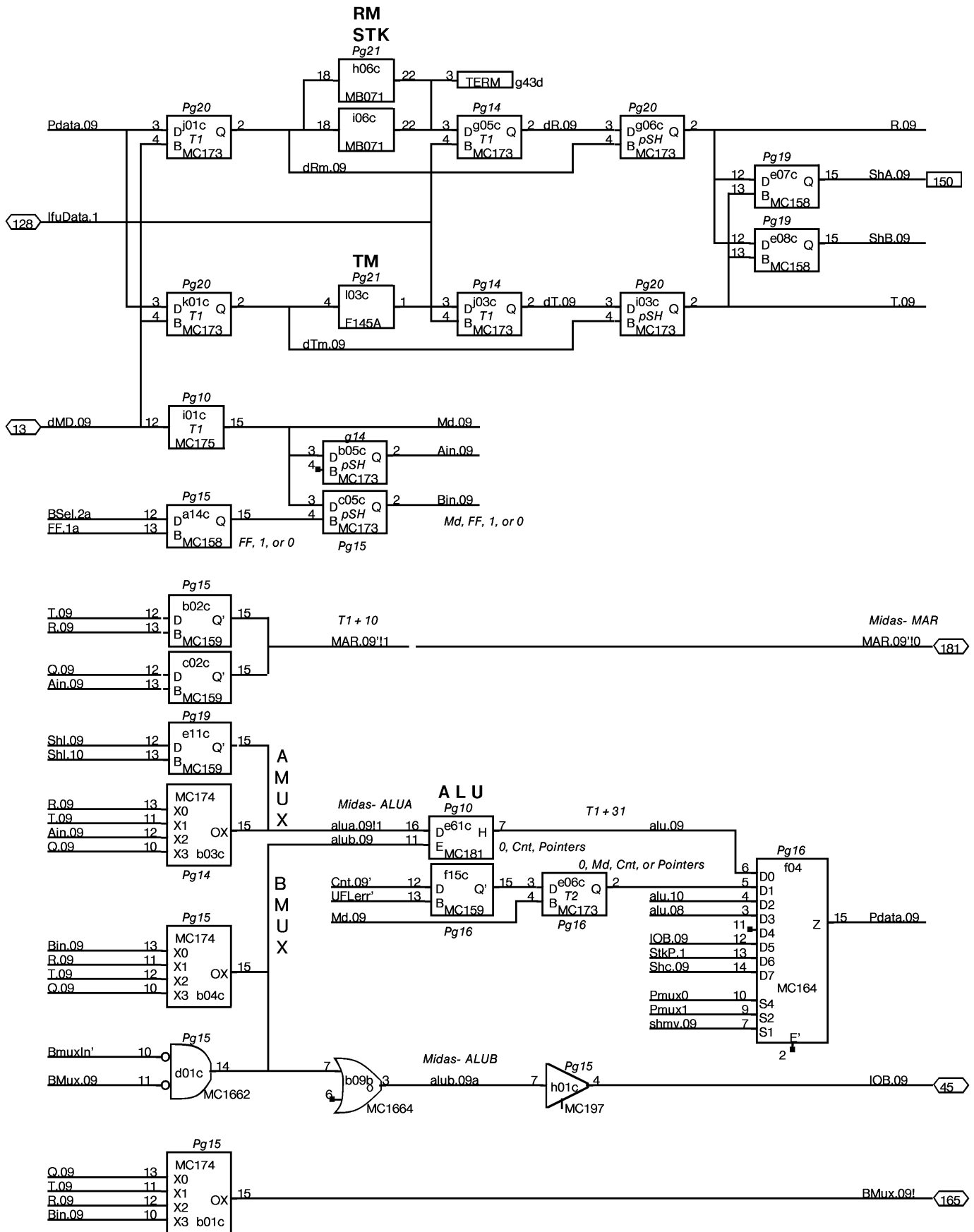
<u>TITLE</u>	<u>Page</u>
Block Diagram	01
Bit slice drawings	02
ALU carry and RM fast-branch	10
ALU function Ram	11
Task & Sub-Task logic	12
Function Field decoding	13
"A" multiplexor & Mar multiplexor control	14
"B" multiplexor control	15
"Processor" multiplexor control	16
Q and Count Registers	17
Shift Control register	18
Shifter multiplexors	19
Rm & T write control	20
Rm, Stk, and T address lines	21
Rm Address control & Bypass	22
Stk Address control & Stk Ufl/Ovfl tests	24
Hold Simulator and Temp Sence	26
Parity logic	27
Midas control and multiplexors	28
Clock distribution	30
Layout	31
Loading Information	32

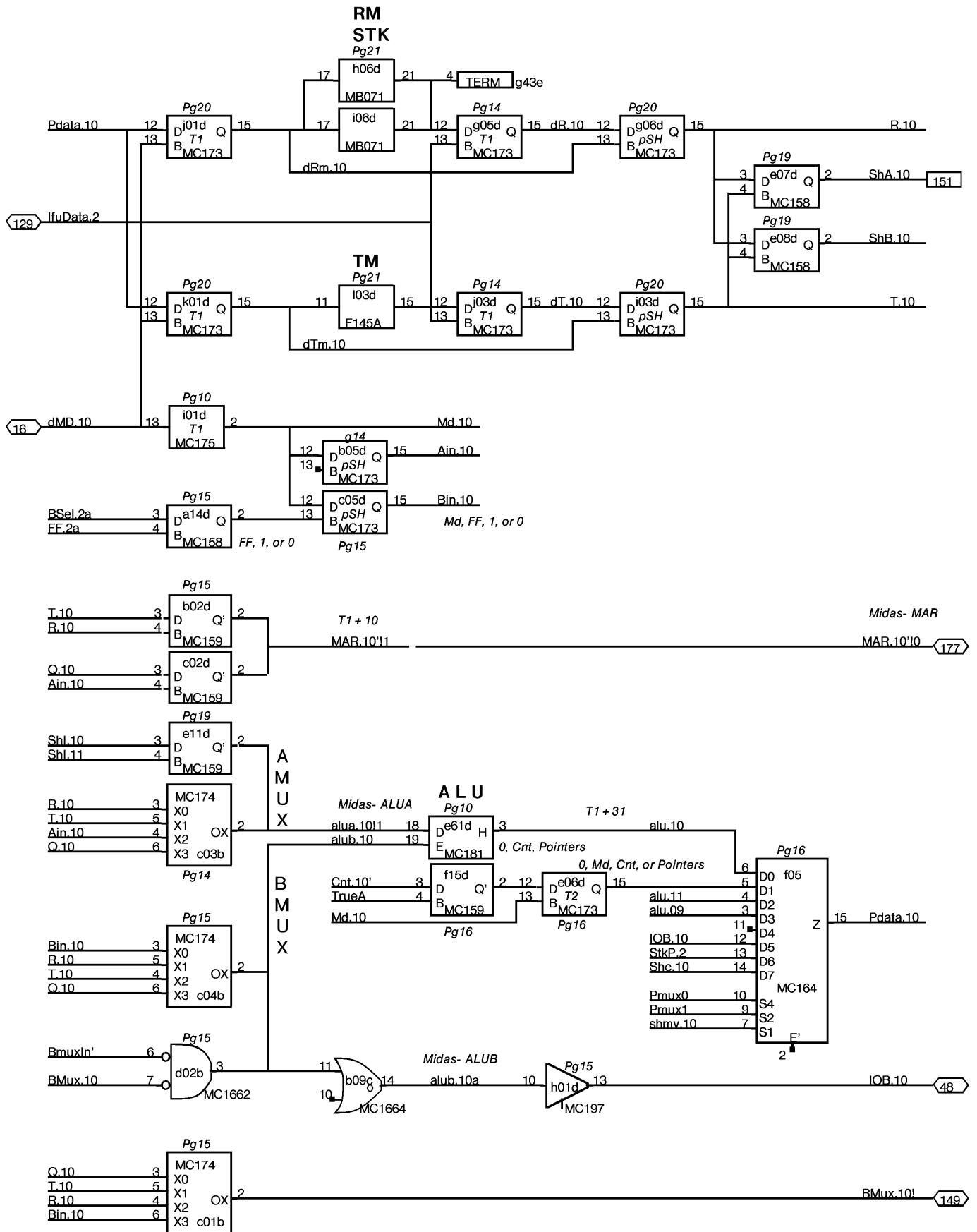


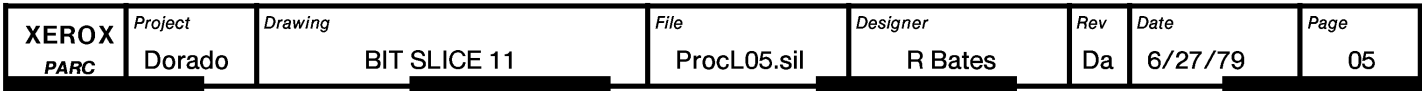
Note: all busses 16 wide unless noted otherwise

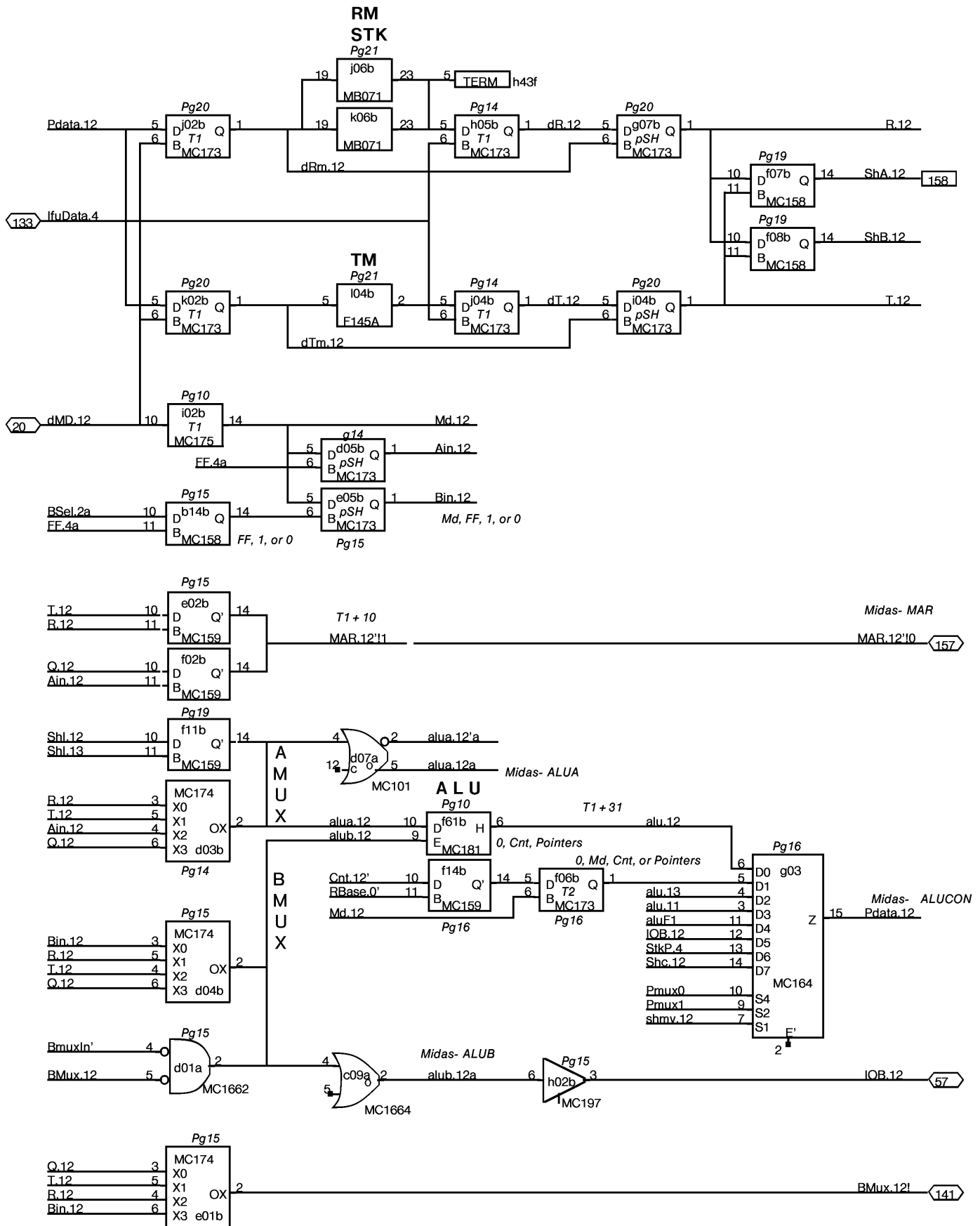
XEROX	Project	Reference	File	Designer	Rev	Date	Page
PARC	Dorado	Block Diagram	ProcL01.sil	R Bates	Da	6/27/79	01

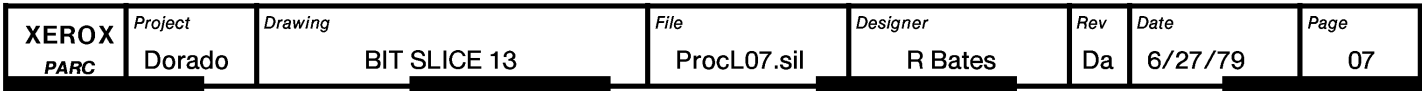


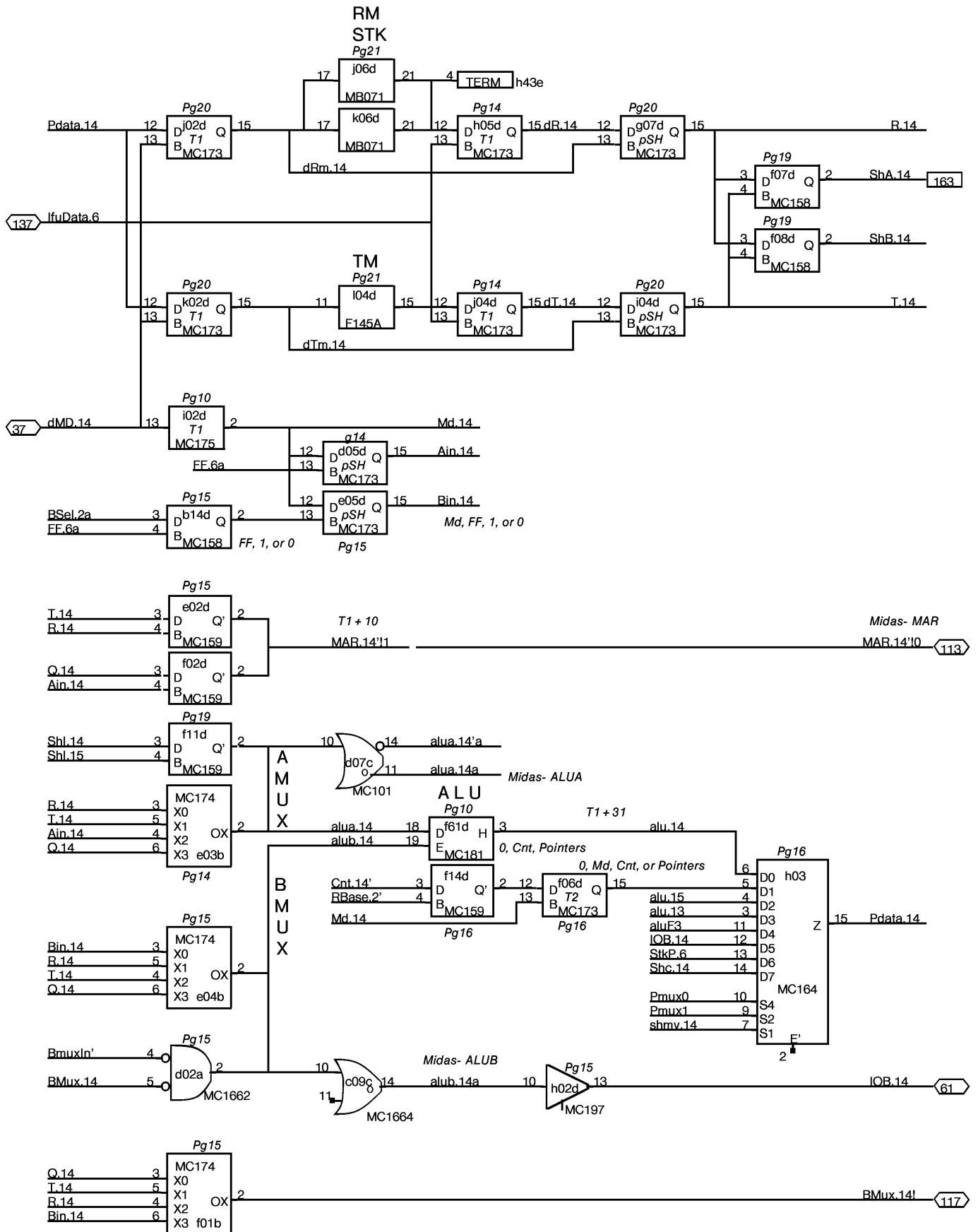


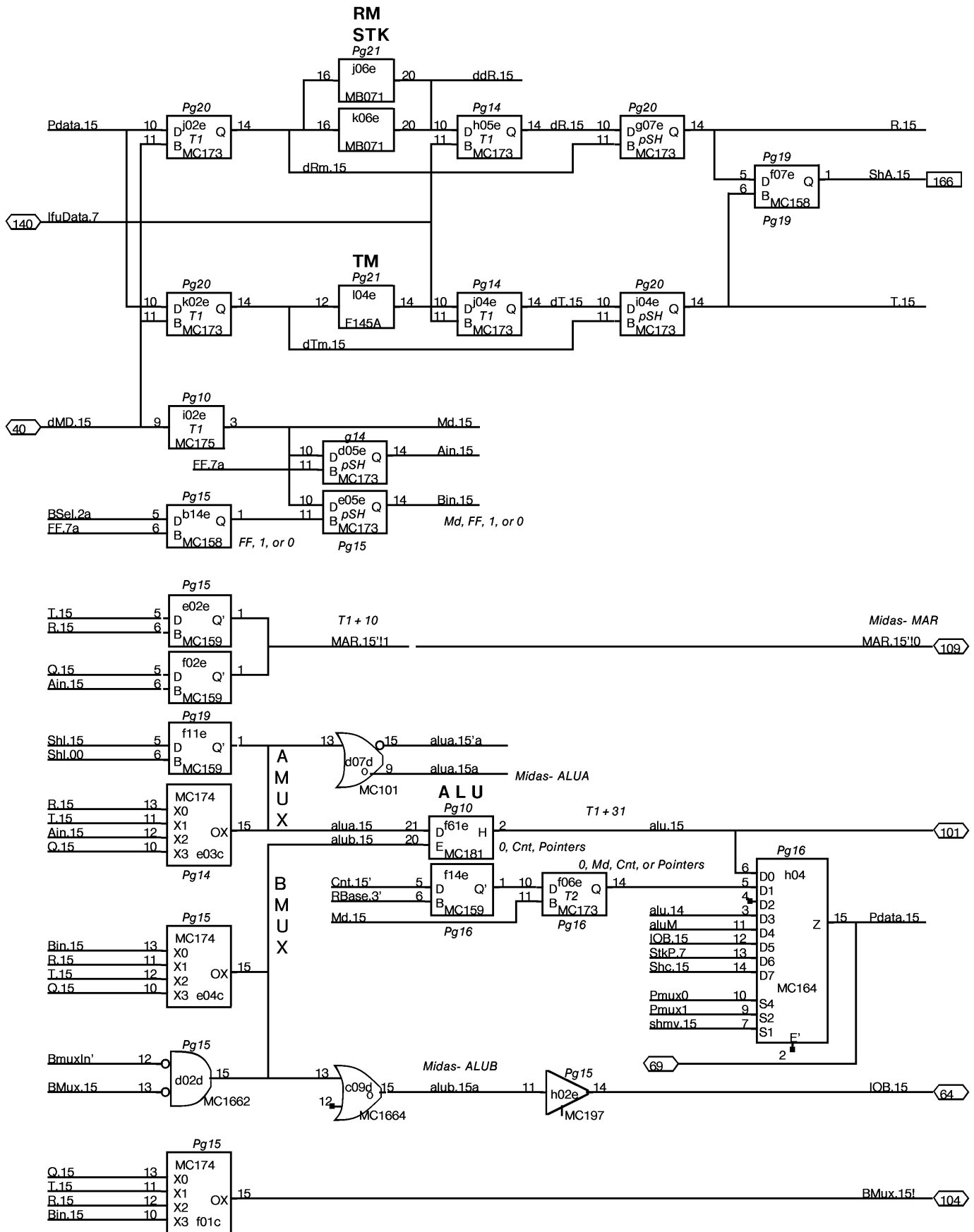










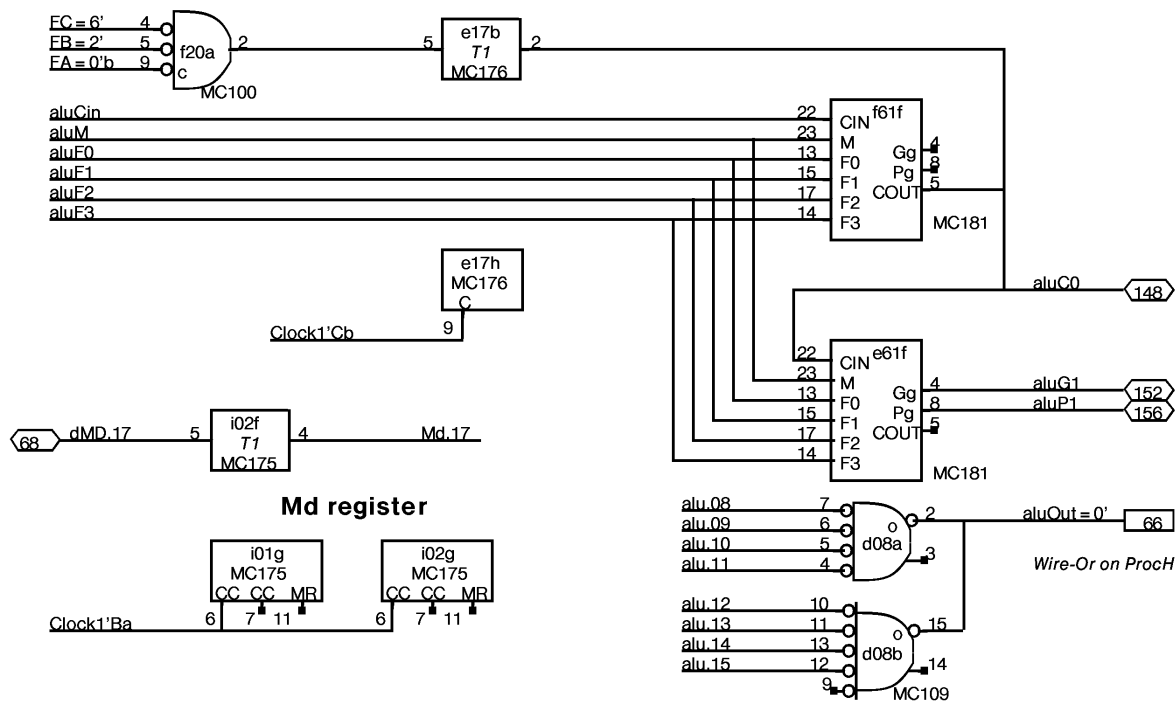


ALU delays

Logical function to output = 11.9
 Arithmetic operation to data = 20.0
 Arithmetic operation to carry = 17.9

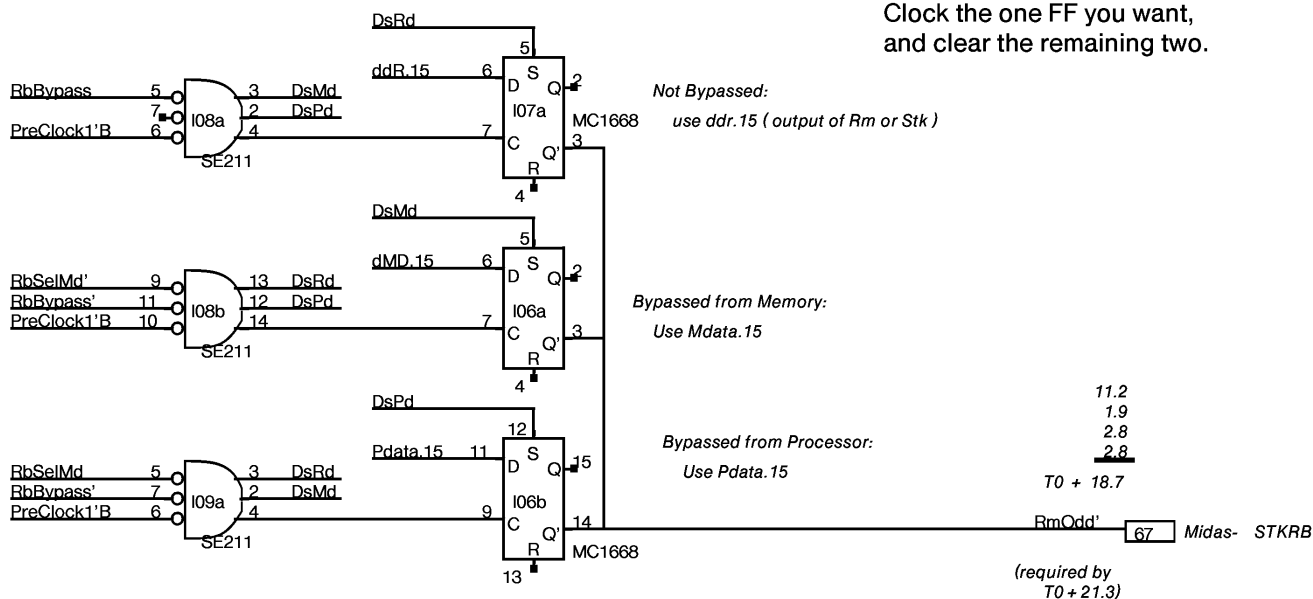
ALU output (assuming 10.2 ns to output of BMux)

Logical Function = 22.0
 Arithmetic operation to data = 30.1
 Arithmetic operation to carry = 28.0



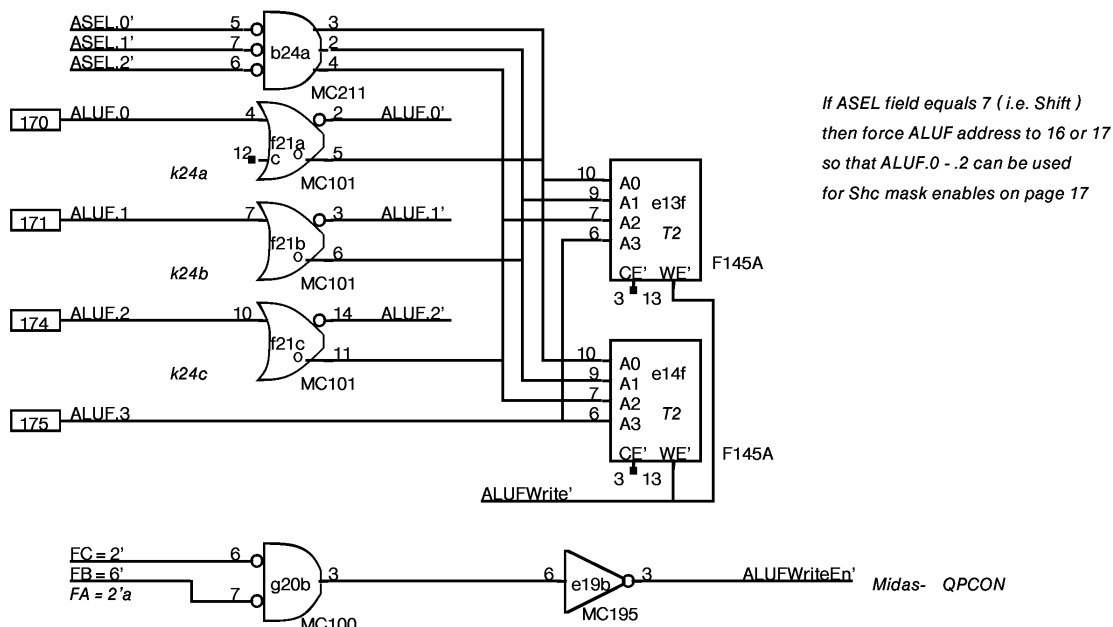
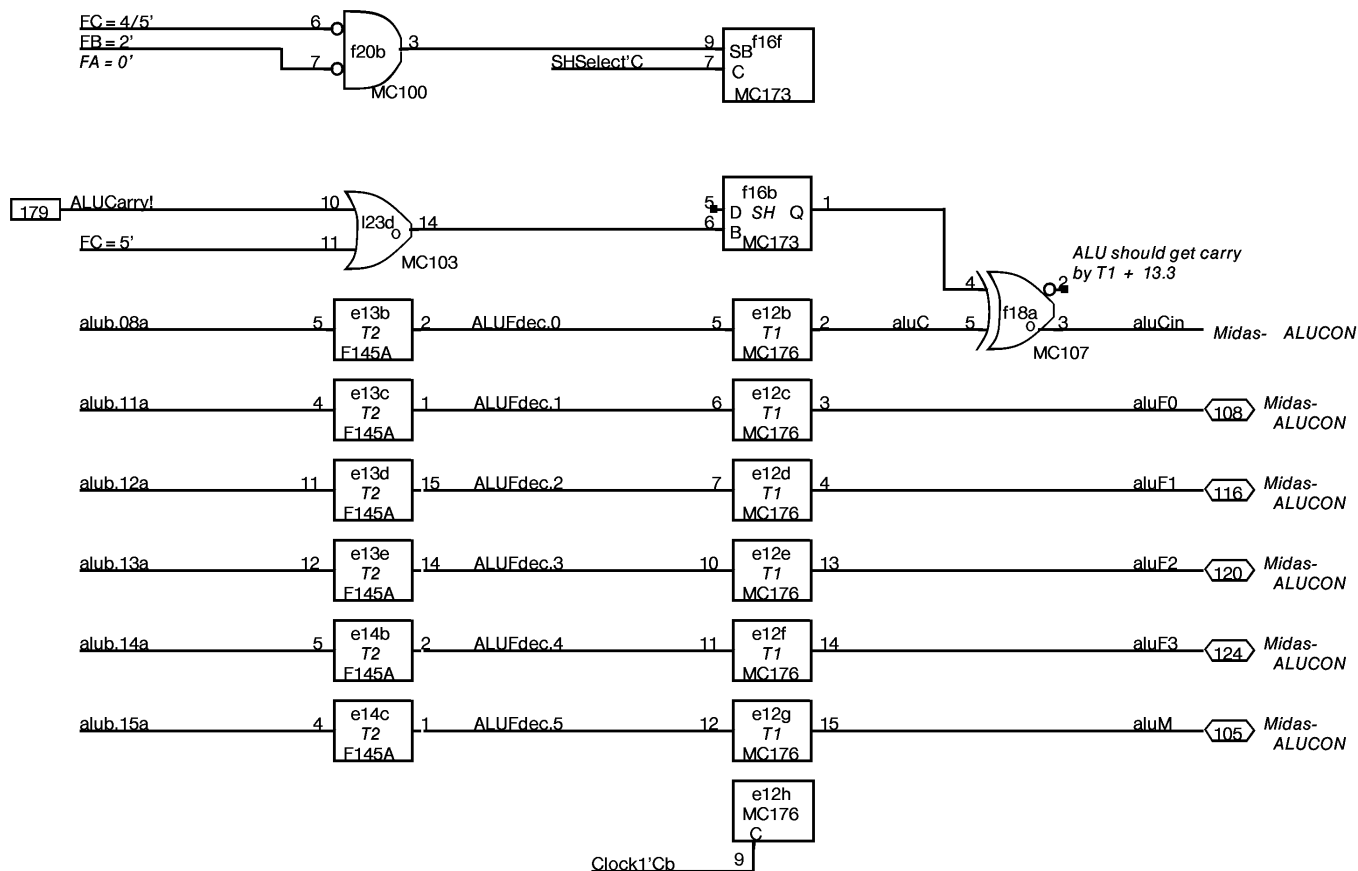
Fast Branch on Rm odd

Clock the one FF you want,
 and clear the remaining two.



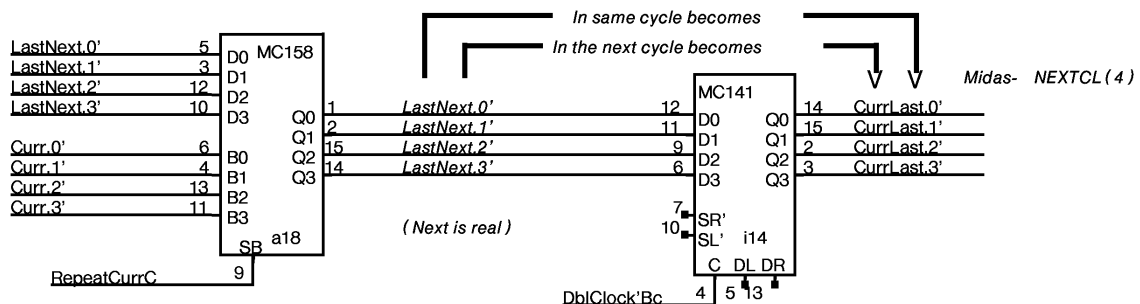
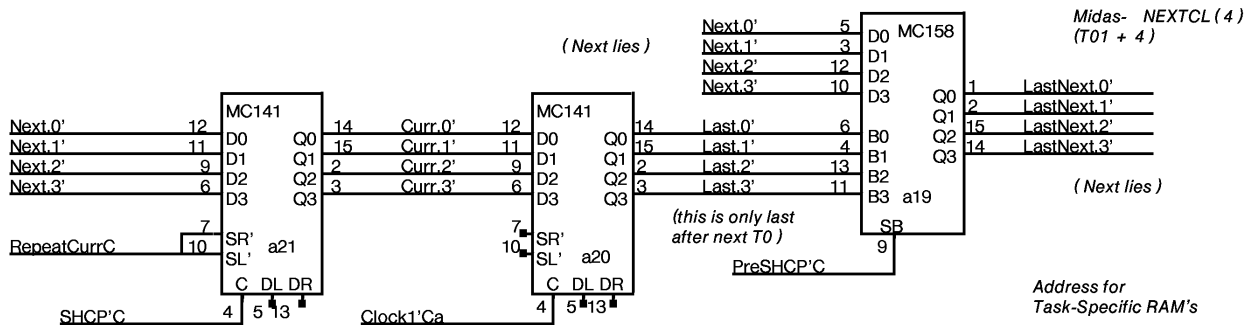
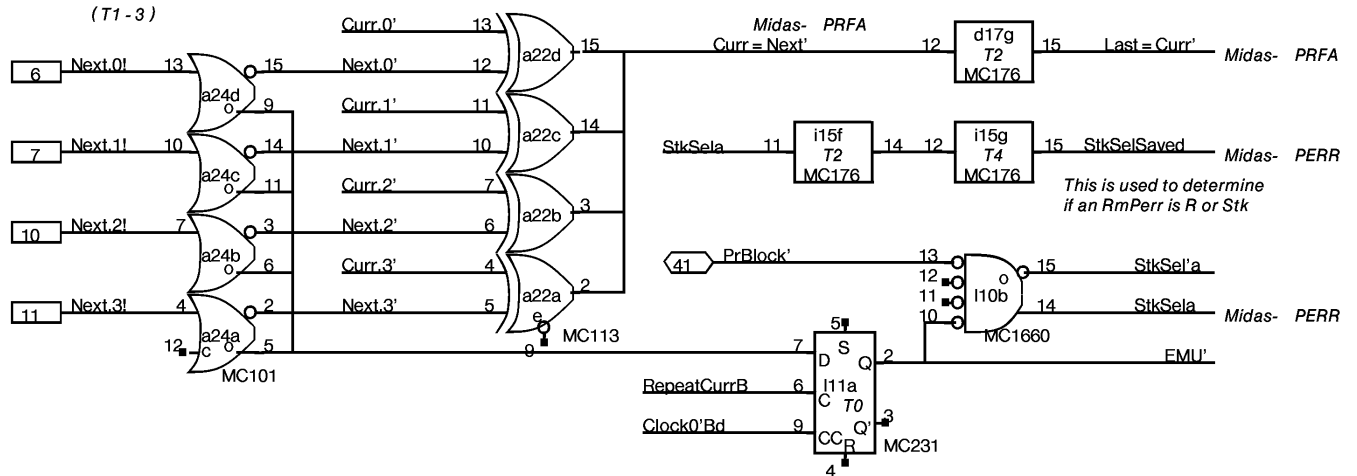
This circuit will correctly
 bypass R from Pdata or Mdata
 When "RisIfData" is in effect, the fast branch
 will be based on the contents of the addressed
 RM or STK, bypassed if necessary.

FF = Cin xor 1
or
FF = Cin xor Cout



If ASEL field equals 7 (i.e. Shift)
then force ALUF address to 16 or 17
so that ALUF.0 - .2 can be used
for Shc mask enables on page 17

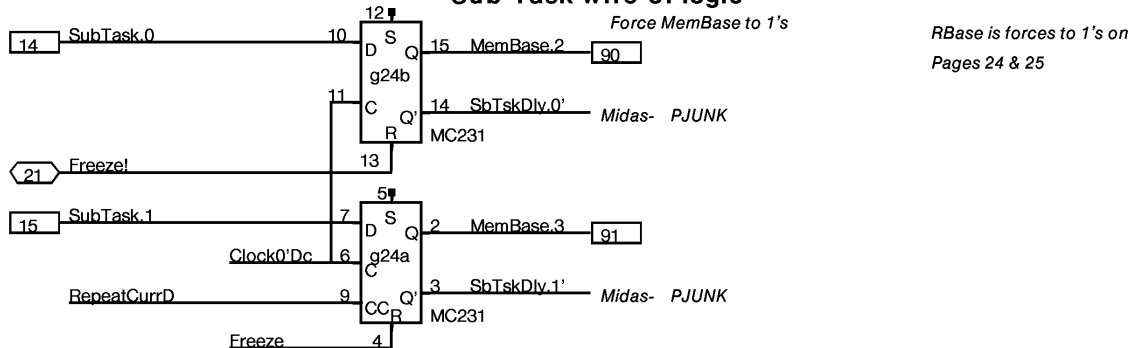
NOTE: Moving k24 to f21 saves 0.4 ns



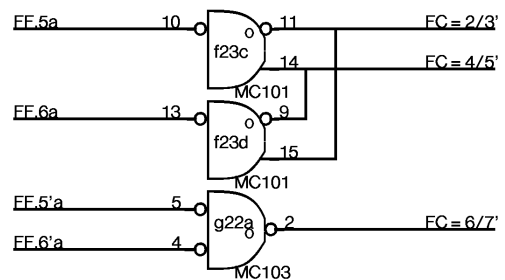
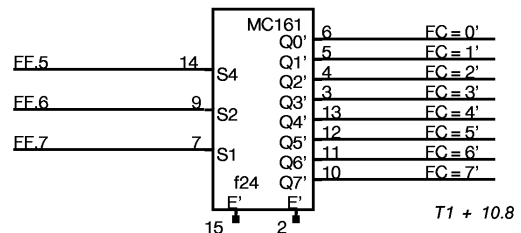
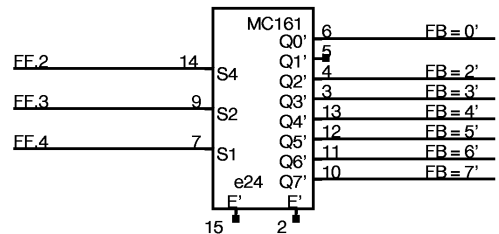
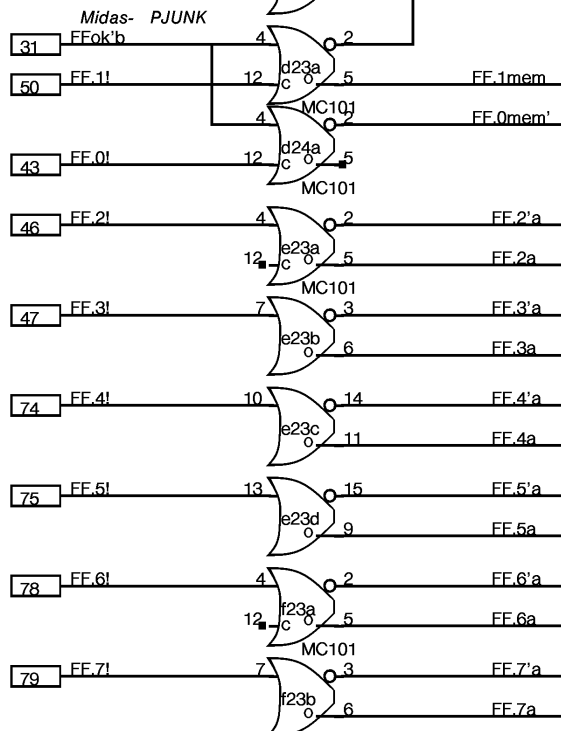
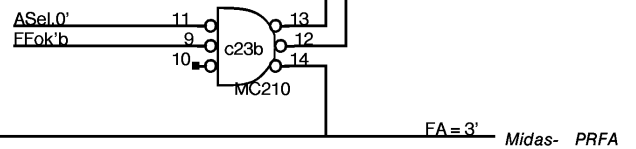
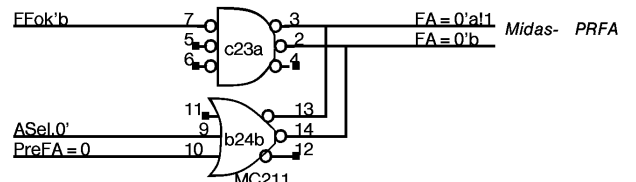
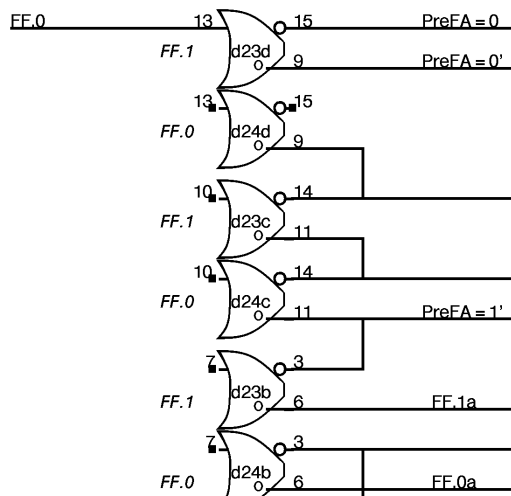
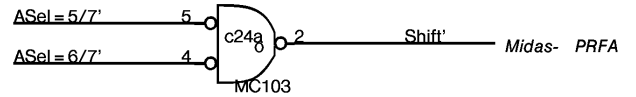
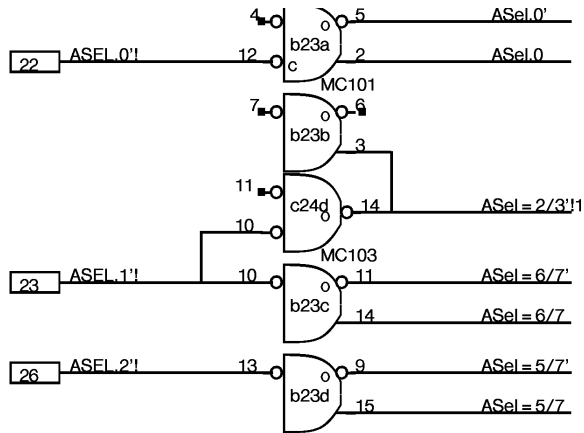
"RepeatCurrent" is asserted on certain occasions when the value on the NEXT bus may be invalid ("Next Lies") due to the combination of Block and Hold.

Task number tracking logic

Sub-Task wire-or logic



RBase is forced to 1's on
Pages 24 & 25



AMux decoding

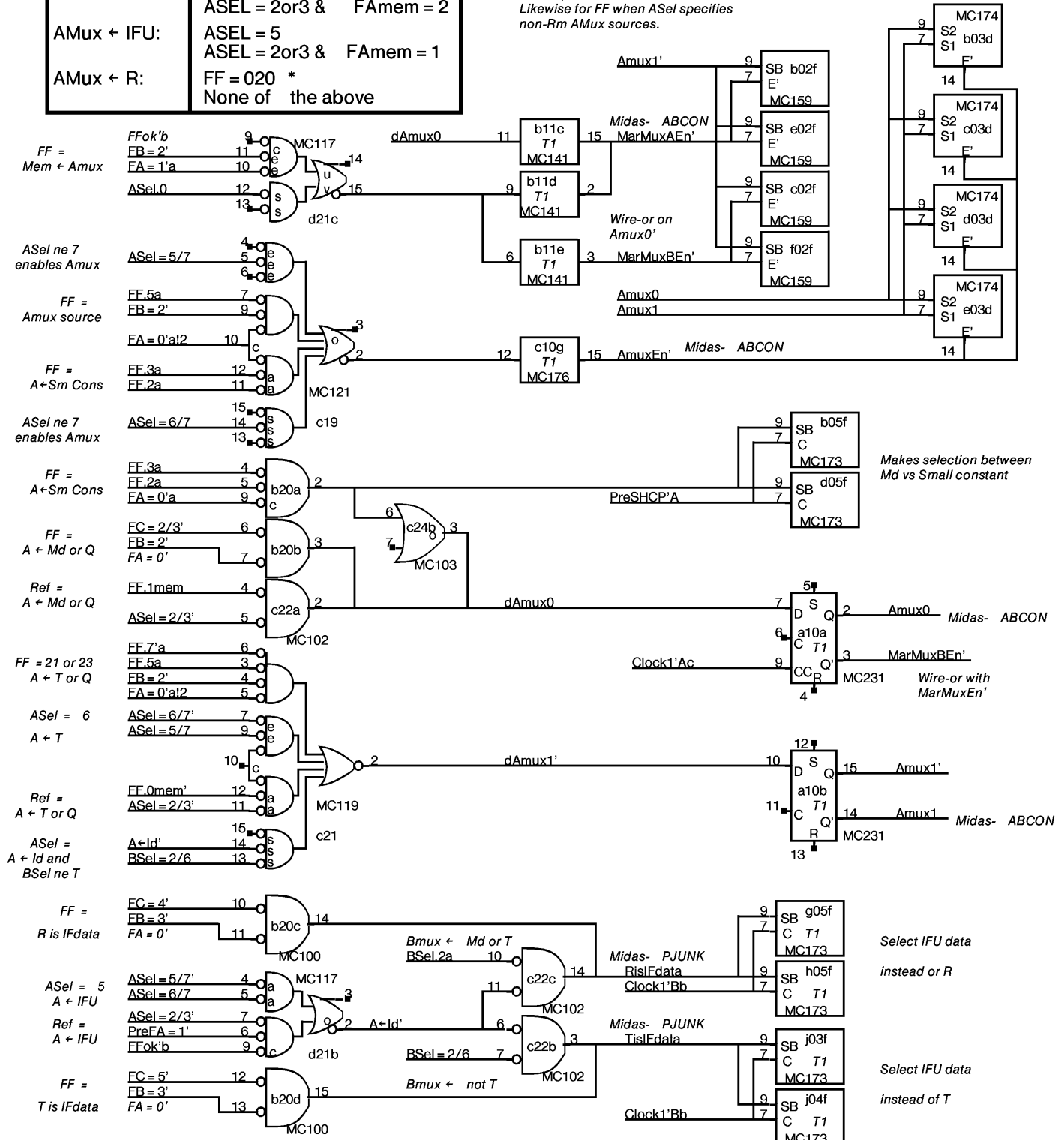
AMux ← FF:	FF = 0-17 *
AMux ← T:	FF = 021 *
	ASEL = 2or3 & FAmem = 3
	ASEL = 6
AMux ← Md:	FF = 022 *
	ASEL = 2or3 & FAmem = 0
AMux ← Q:	FF = 023 *
	ASEL = 2or3 & FAmem = 2
AMux ← IFU:	ASEL = 5
	ASEL = 2or3 & FAmem = 1
AMux ← R:	FF = 020 *
	None of the above

* The Amux is disabled by ASEL = 7 unless one of these codes are in effect

NOTE: ASEL selects and FF selects for the AMux are "OR'd" by this hardware. Thus ASEL codes selecting non-Rm sources of Amux must not be used when an FF specifies an ASEL source. Likewise for FF when ASEL specifies non-Rm AMux sources.

AMux encoding

Mux Input	Source
0	R or IFU data
1	T or IFU data
2	Md or Small Const.
3	Q

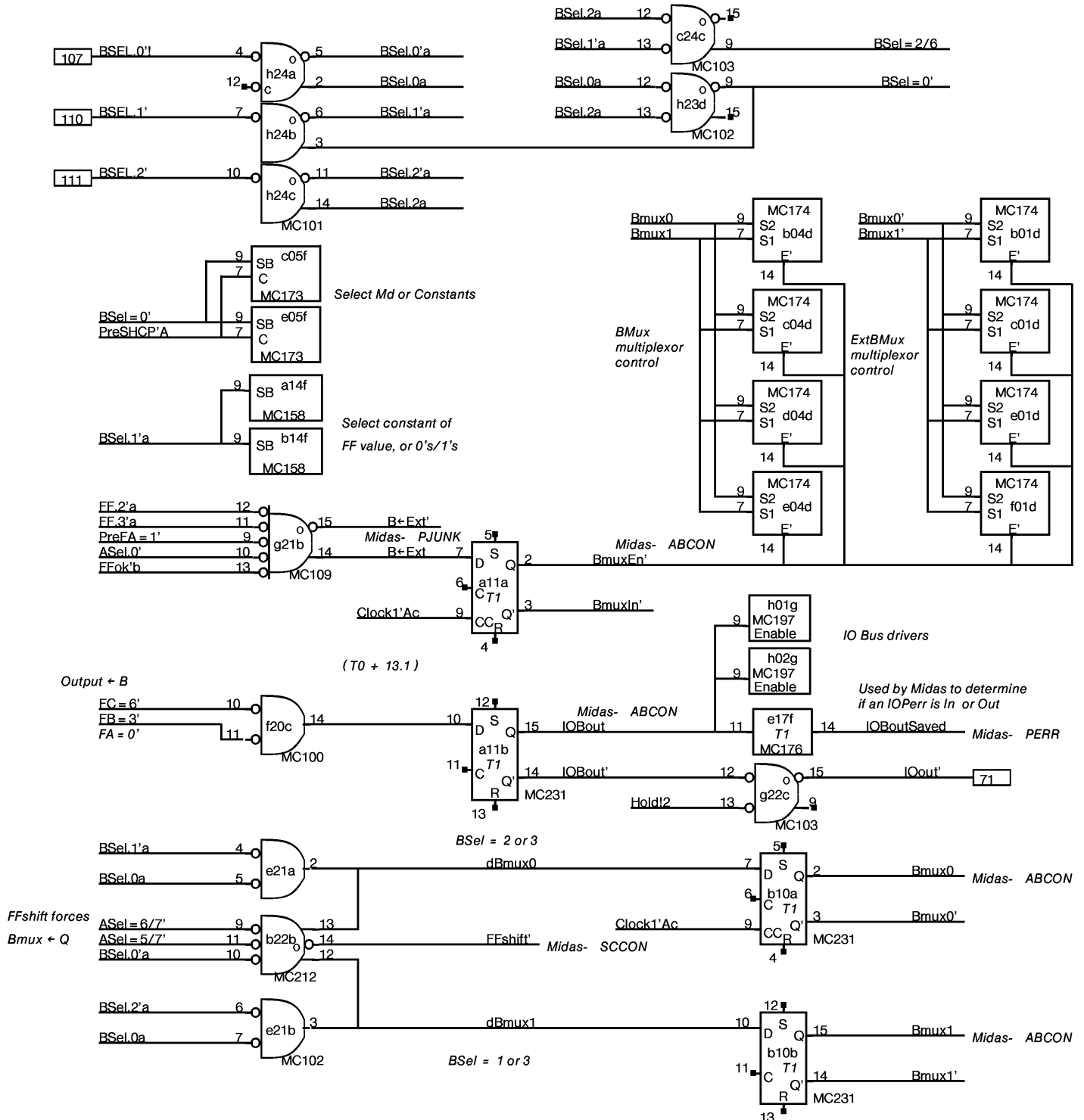


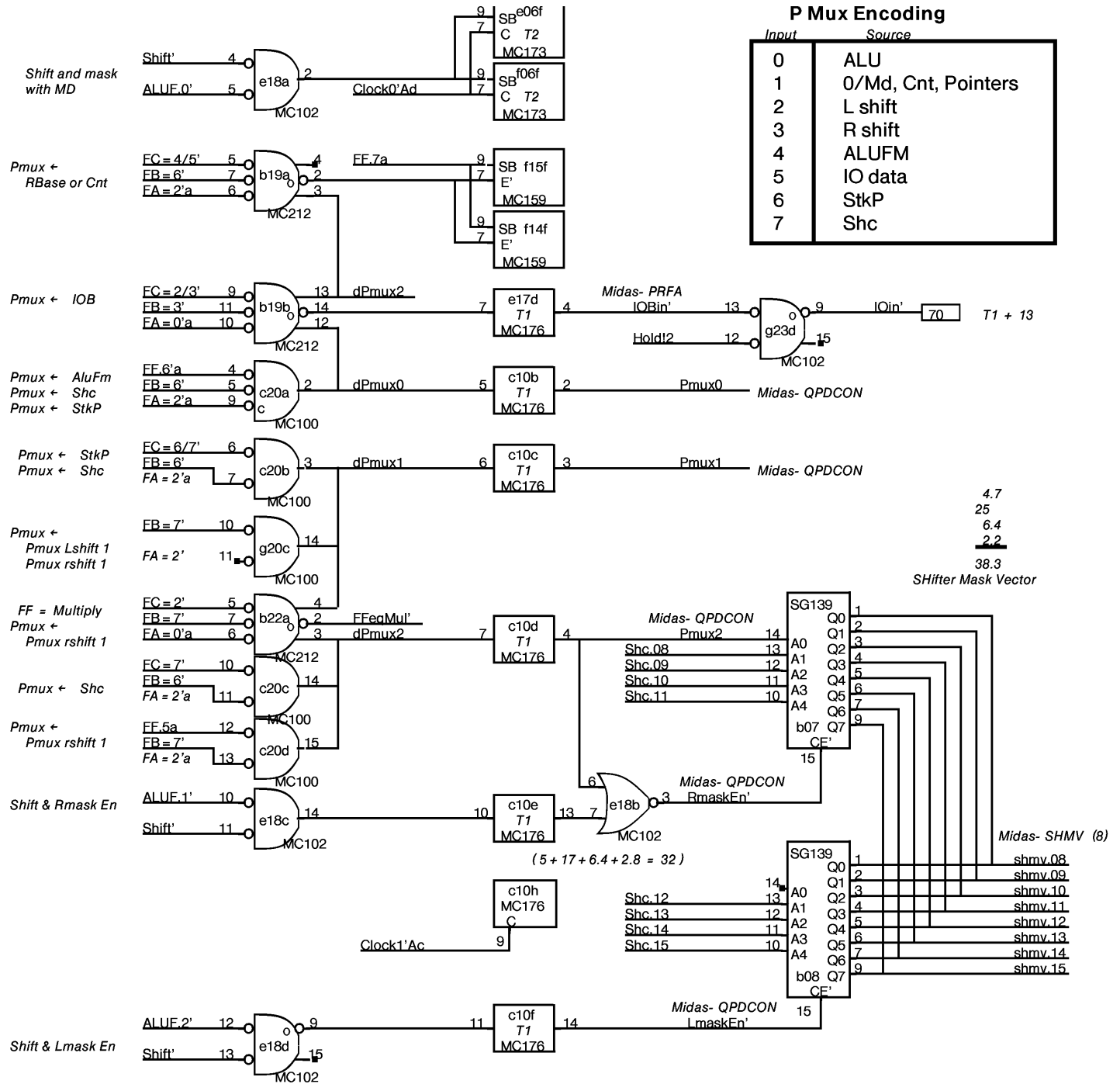
BSEL field decoding

BSEL	INTERNAL	EXTERNAL
0	Md	--
1	R	--
2	T	Hold ← B
3	Q	Q ← B
4	0,,FF	--
5	377,FF	--
6	FF,,0	--
7	FF,,377	--

BMux encoding

BMux	
0	Md or Constant
1	R
2	T
3	Q





NOTE: The prom patterns are designed so that a one into address bit 0 will produce all 1's on the output. This allows the odd address inputs to Pmux to be selected.

Lmask

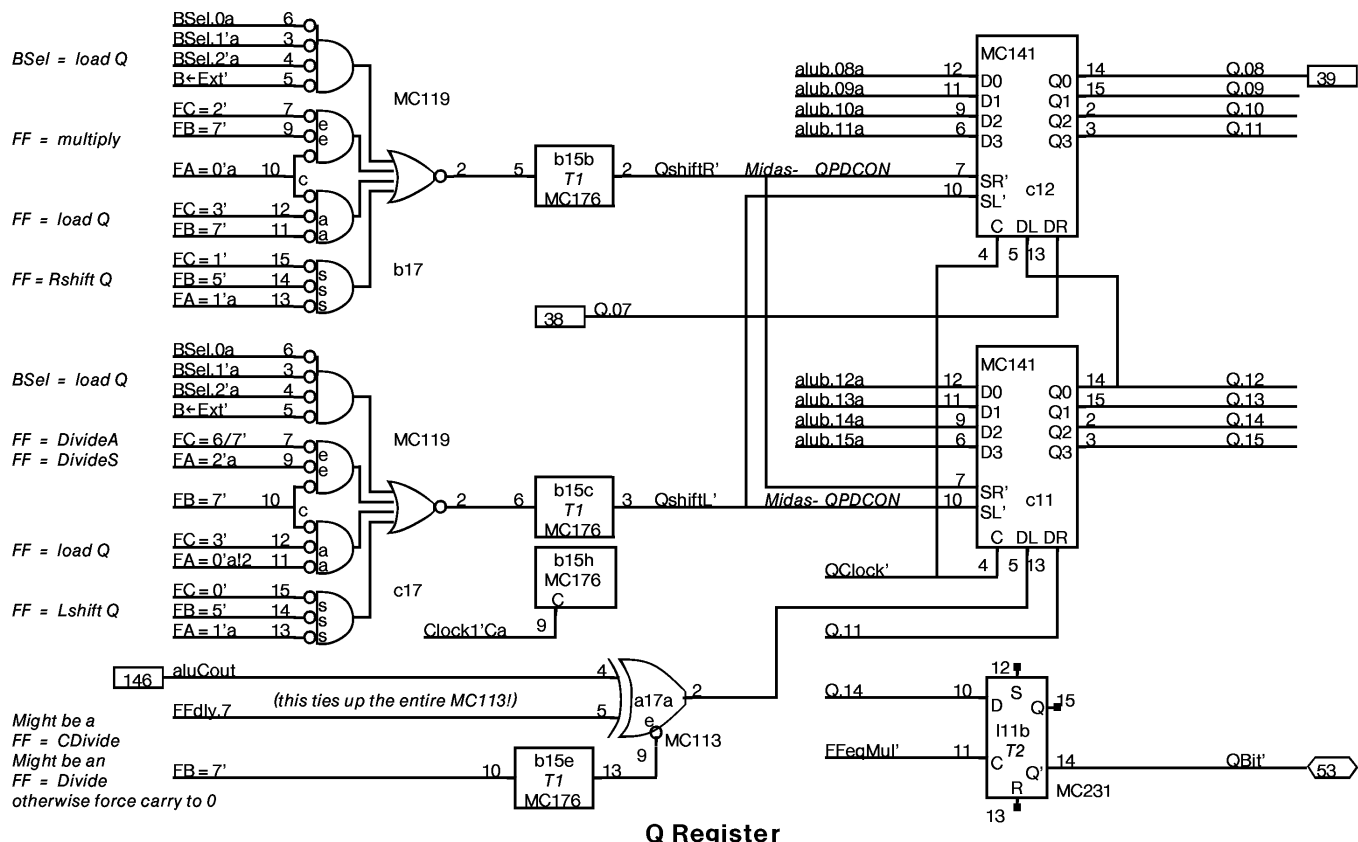
Shc.12-15

	08	09	10	11	12	13	14	15
0	0	0	0	0	0	0	0	0
1	0	0	0	0	0	0	0	0
2	0	0	0	0	0	0	0	0
3	0	0	0	0	0	0	0	0
4	0	0	0	0	0	0	0	0
5	0	0	0	0	0	0	0	0
6	0	0	0	0	0	0	0	0
7	0	0	0	0	0	0	0	0
10	0	0	0	0	0	0	0	0
11	1	0	0	0	0	0	0	0
12	1	1	0	0	0	0	0	0
13	1	1	1	0	0	0	0	0
14	1	1	1	1	0	0	0	0
15	1	1	1	1	1	0	0	0
16	1	1	1	1	1	1	0	0
17	1	1	1	1	1	1	1	0
20- 37	1	1	1	1	1	1	1	1

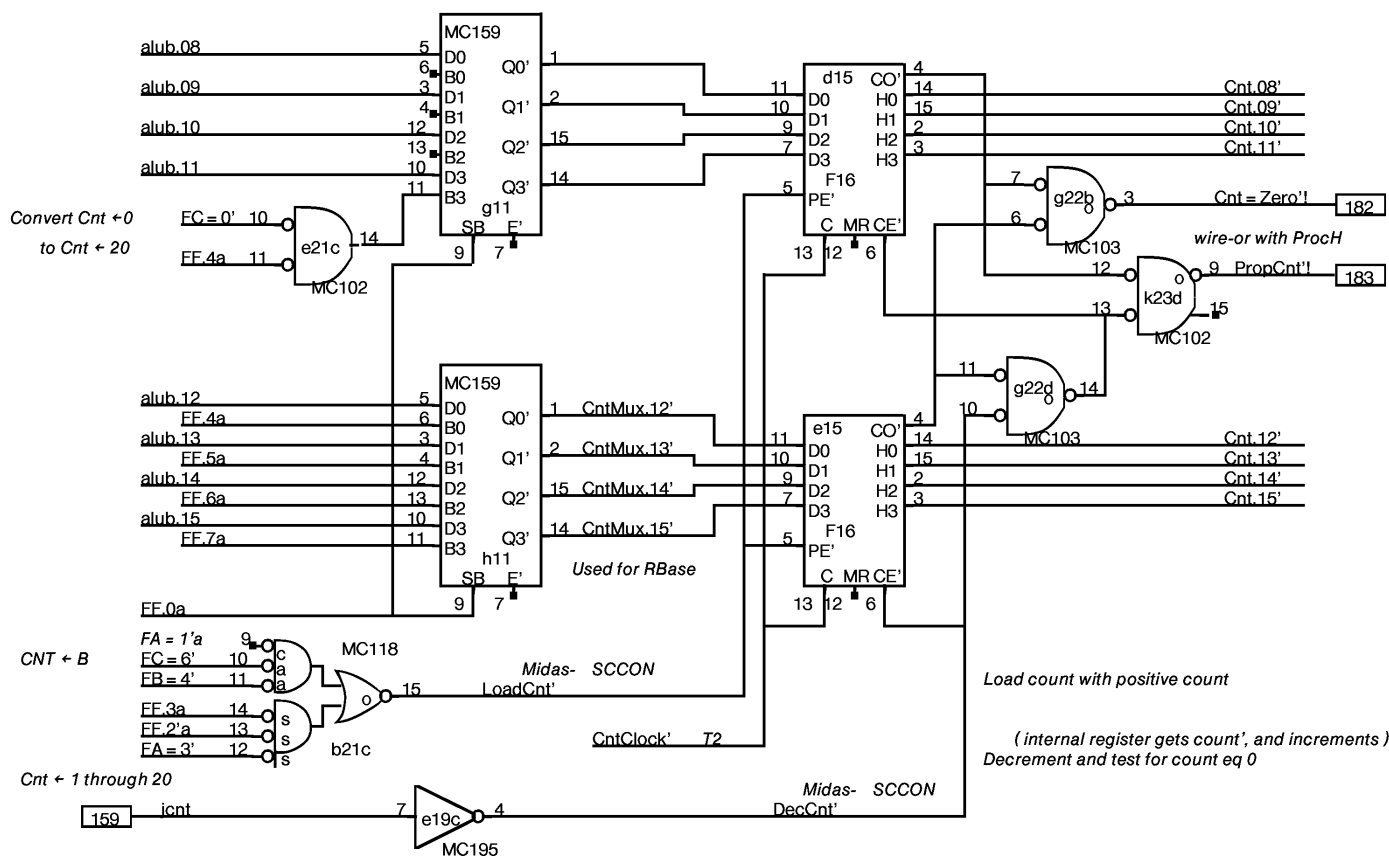
Rmask

Shc.08-11

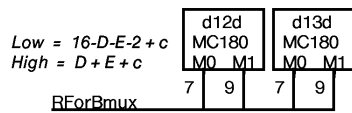
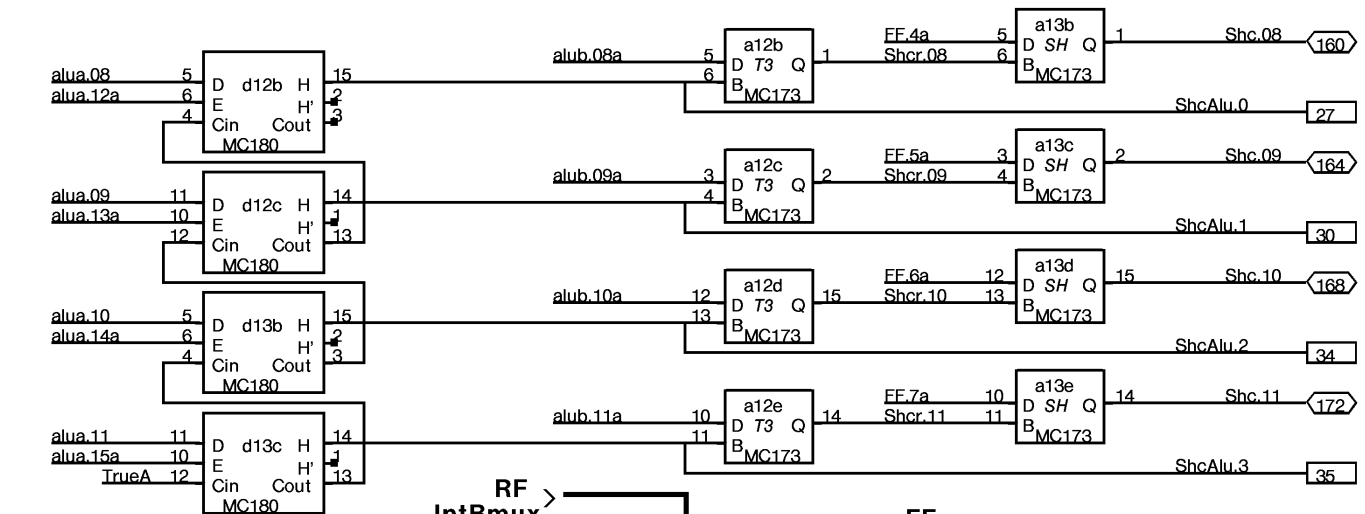
	08	09	10	11	12	13	14	15
0	0	0	0	0	0	0	0	0
1	0	0	0	0	0	0	0	1
2	0	0	0	0	0	0	0	1
3	0	0	0	0	0	0	1	1
4	0	0	0	0	0	1	1	1
5	0	0	0	1	1	1	1	1
6	0	0	1	1	1	1	1	1
7	0	1	1	1	1	1	1	1
10	1	1	1	1	1	1	1	1
11	1	1	1	1	1	1	1	1
12	1	1	1	1	1	1	1	1
13	1	1	1	1	1	1	1	1
14	1	1	1	1	1	1	1	1
15	1	1	1	1	1	1	1	1
16	1	1	1	1	1	1	1	1
17	1	1	1	1	1	1	1	1
20- 37	1	1	1	1	1	1	1	1



Count Register



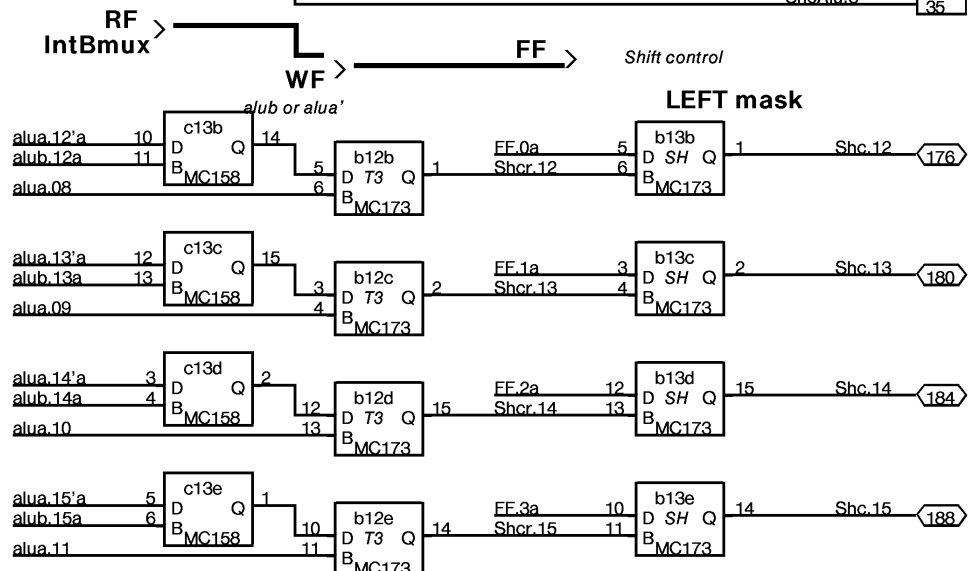
RIGHT mask



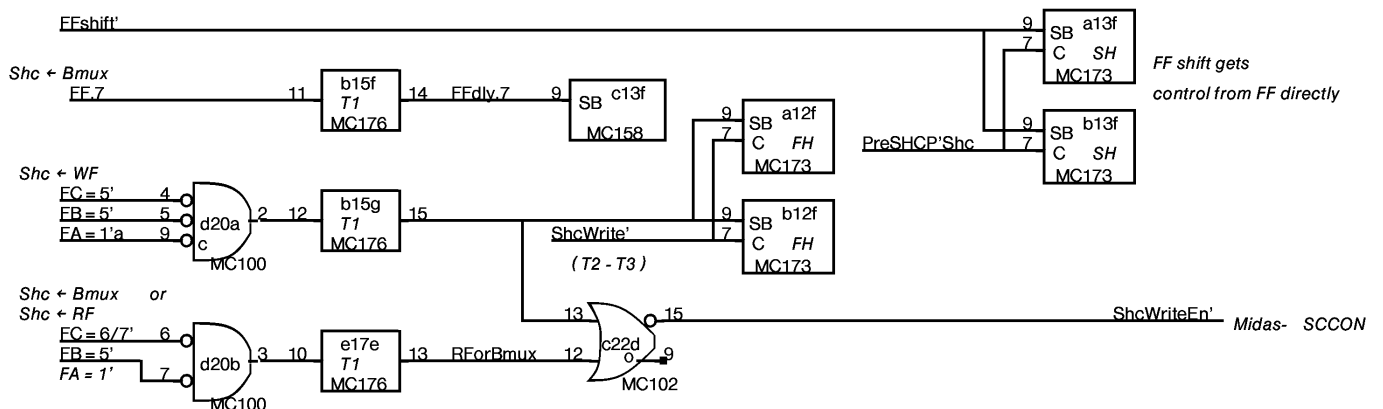
These alu's perform

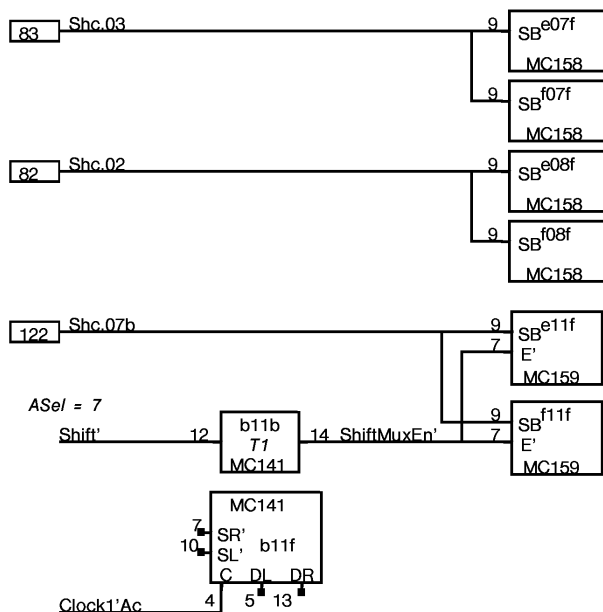
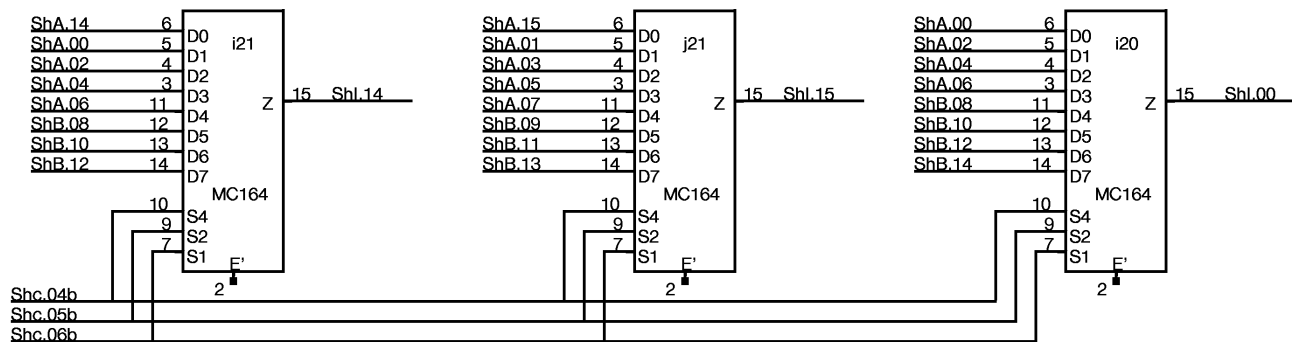
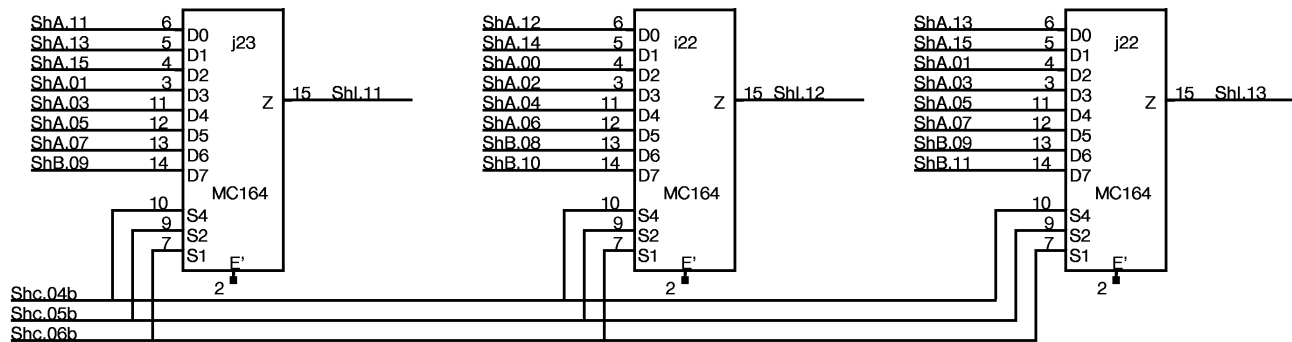
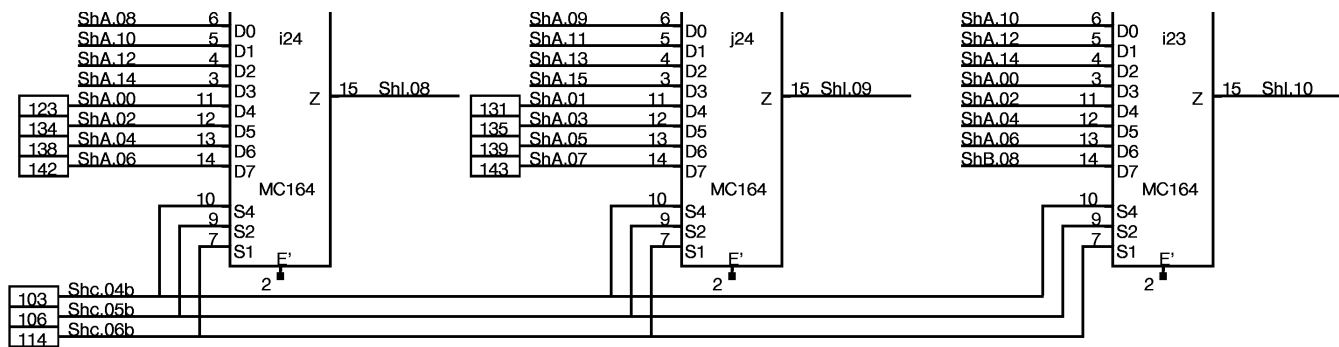
	RF	WF
Shift Count	$P + S + 1$	$16 - P - S - 1$
Right Mask	Don't Care	$16 - P - S - 1$
LeftMask	$16 - S - 1$ (ie S')	P

"P" = 8:11
"S" = 12:15

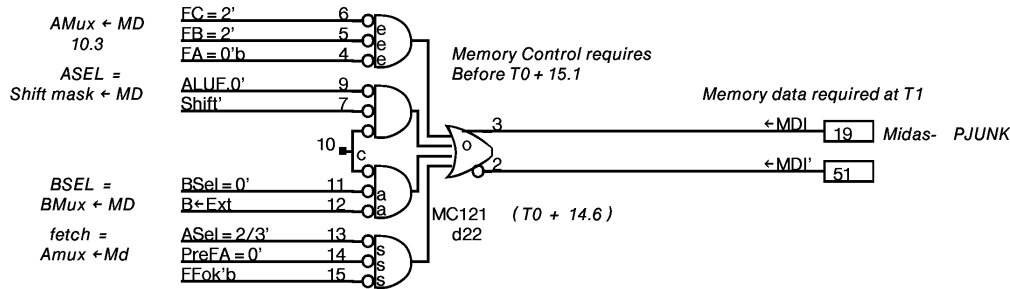


Done this way so EF/IF shifts
do not use Shc - IO tasks can
use the shifter without saving
and restoring Shc.



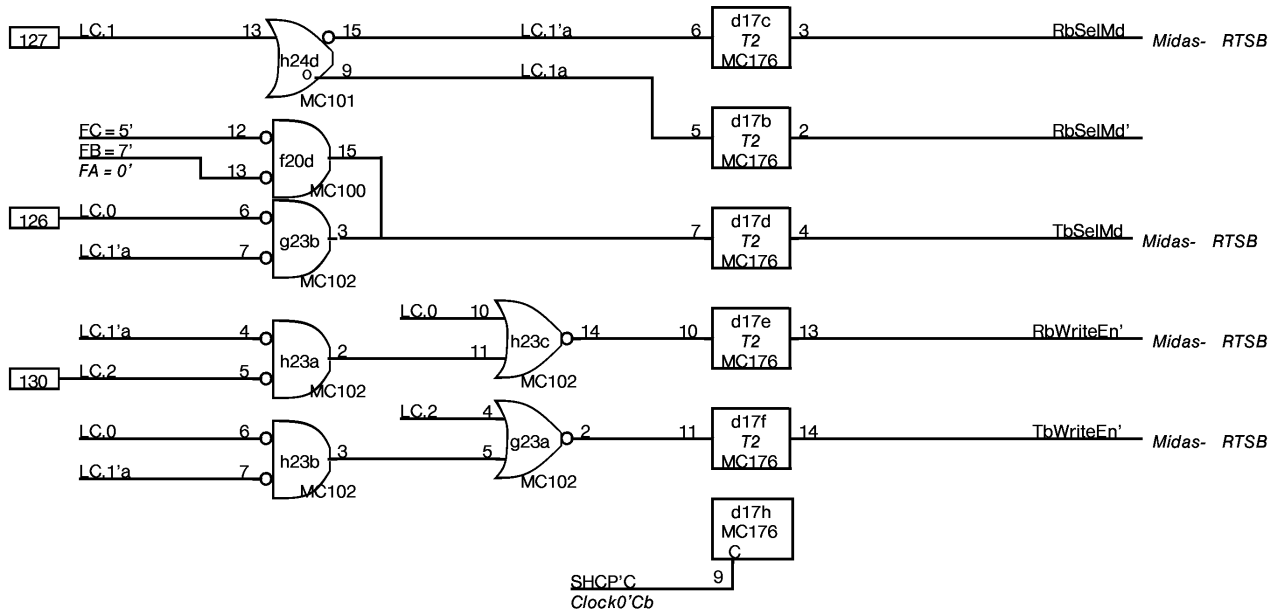
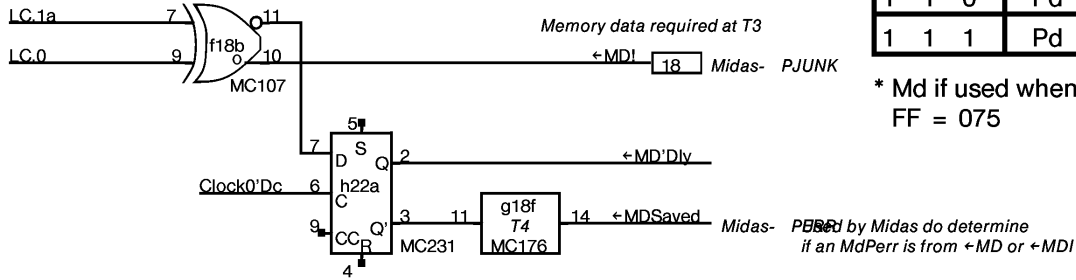


Shc.02	Shc.03	Shc.04 - .07
0	0	16 bit Left cycle of R
0	1	16 bit Left cycle of R,,T
1	0	16 bit Left cycle of T,,R
1	1	16 bit Left cycle of T

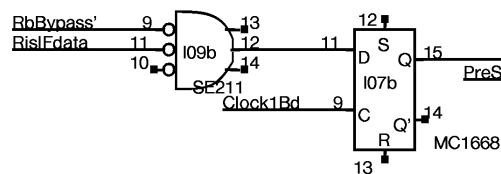
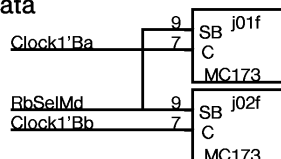


LC			R	T
0	1	2		
0	0	0	-	-
0	0	1	-	Pd *
0	1	0	Pd	Md
0	1	1	-	Md
1	0	0	Md	-
1	0	1	Md	Pd *
1	1	0	Pd	-
1	1	1	Pd	Pd *

* Md if used when
FF = 075

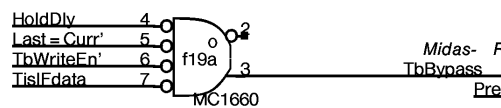
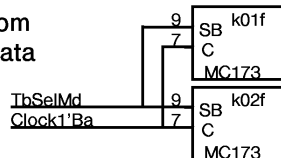


Write R From
Pdata or Mdata

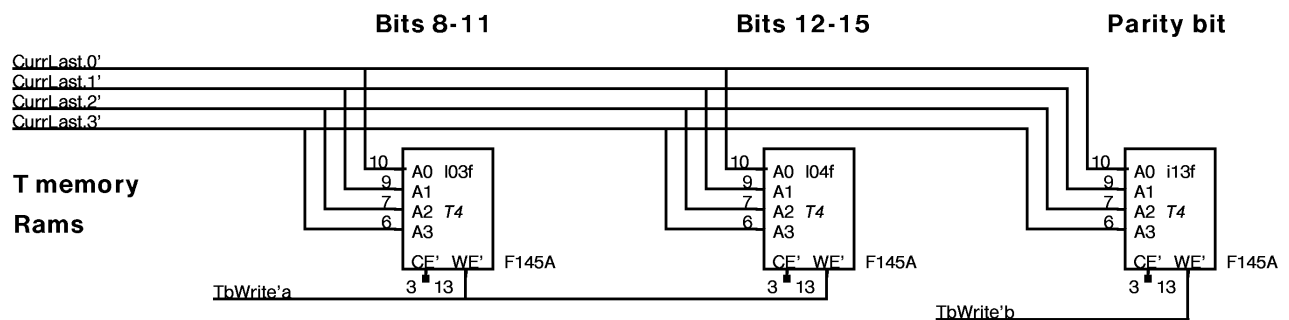
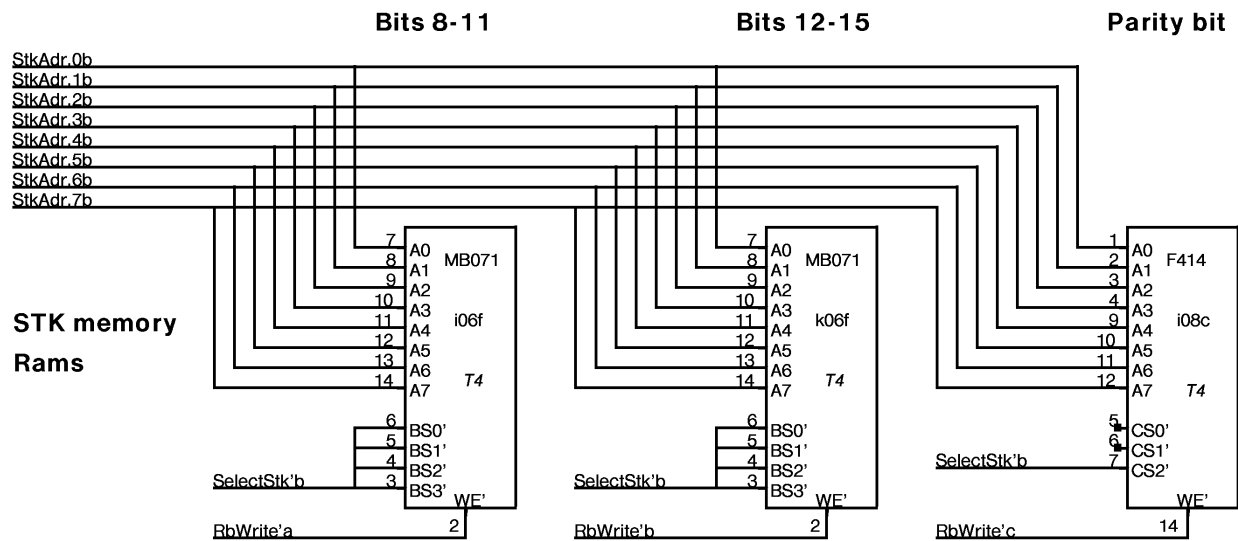
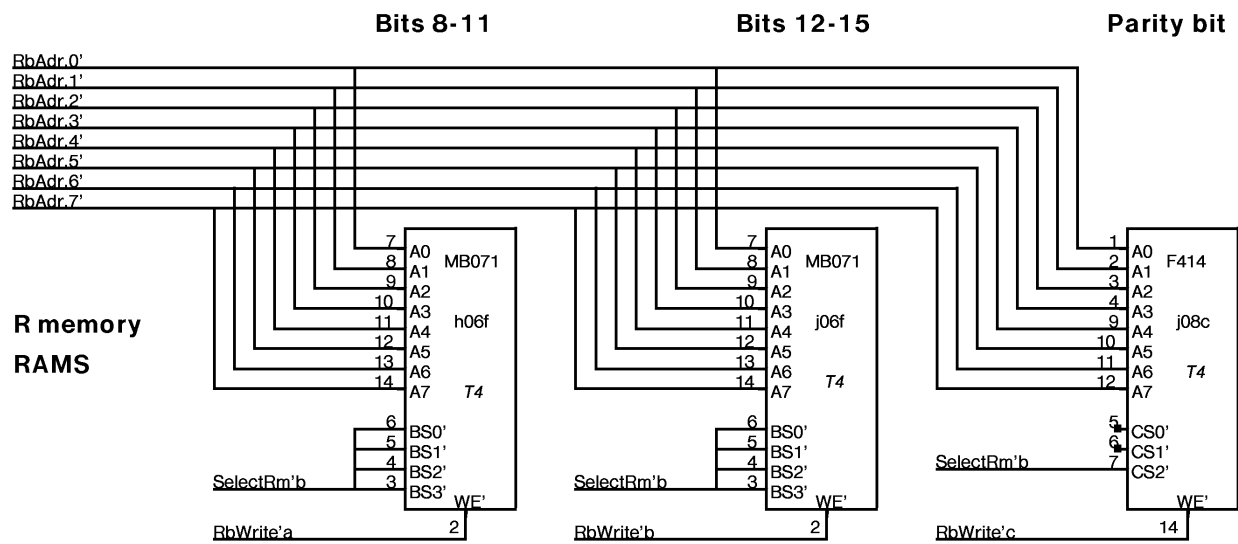


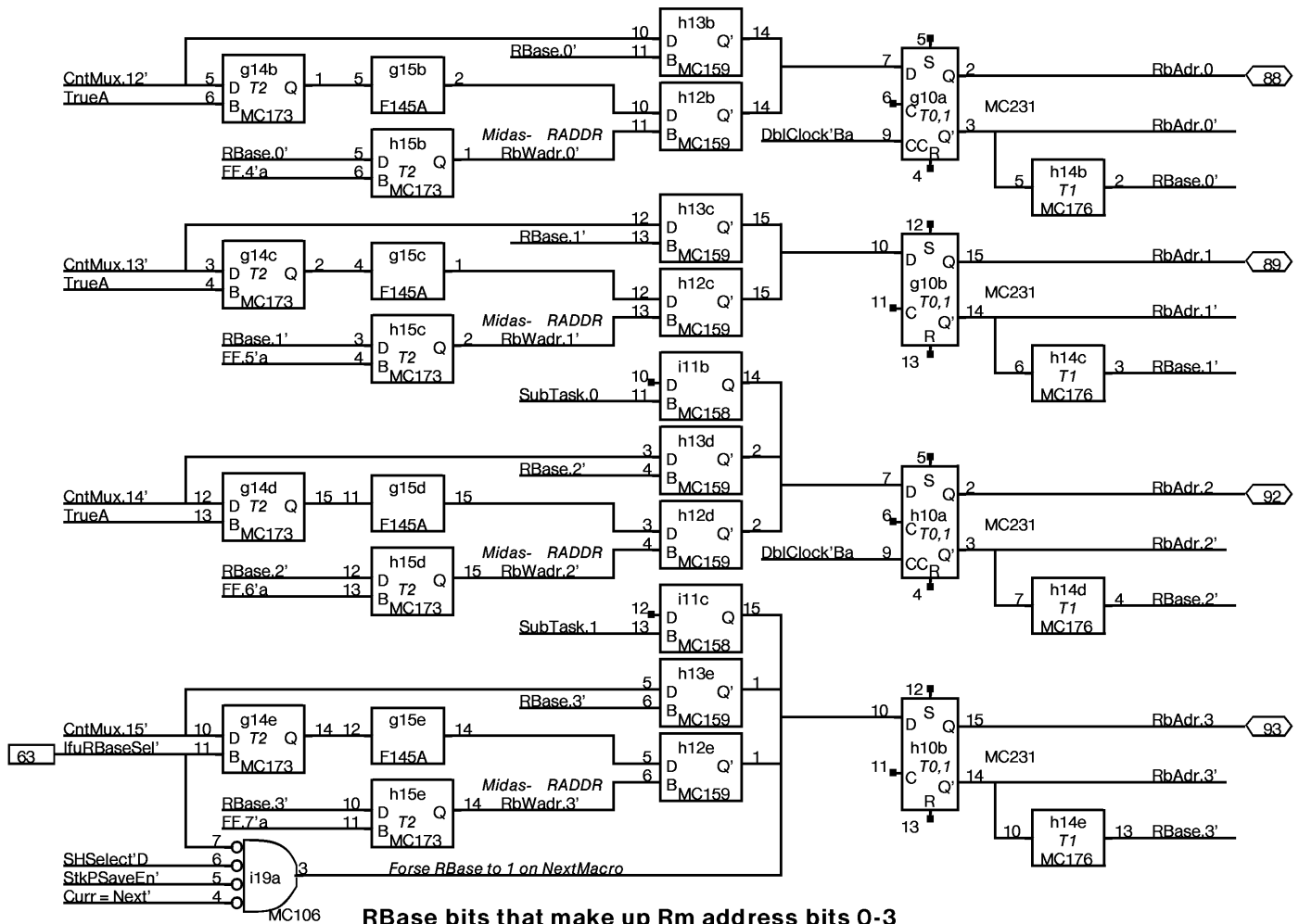
Bypass R from
Pdata or Mdata

Write T From
Pdata or Mdata



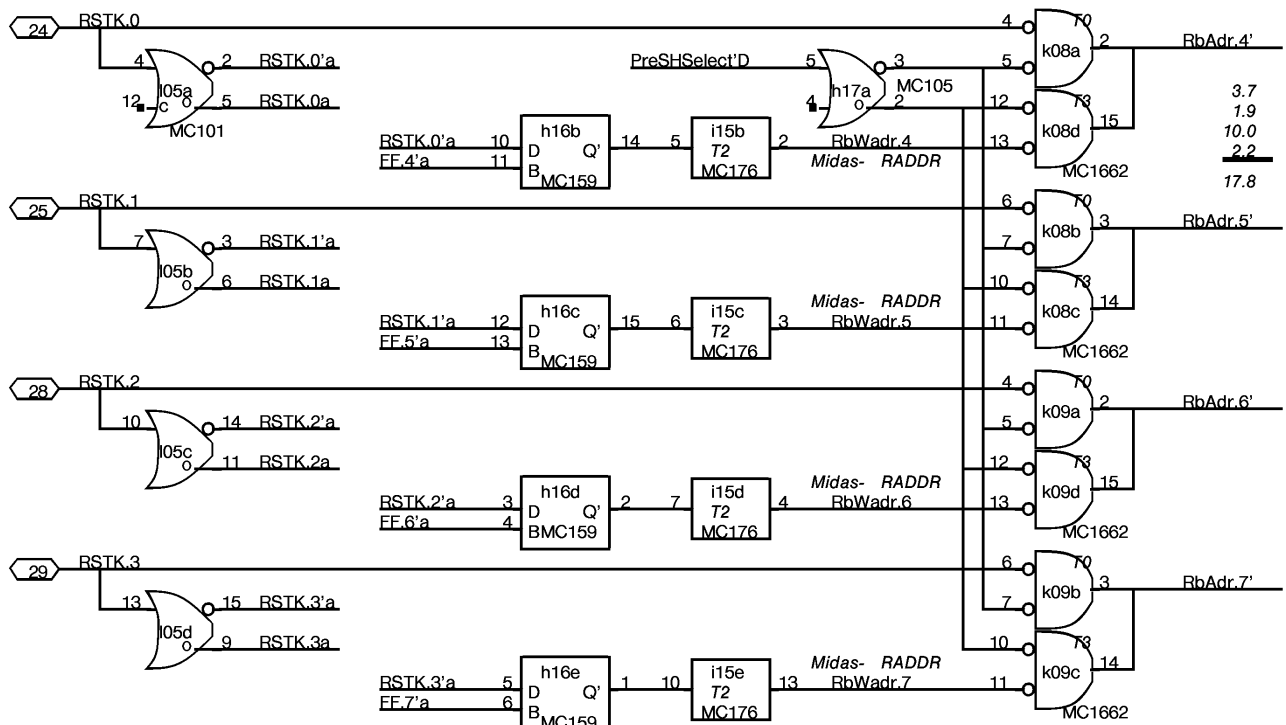
Bypass T from
Pdata or Mdata

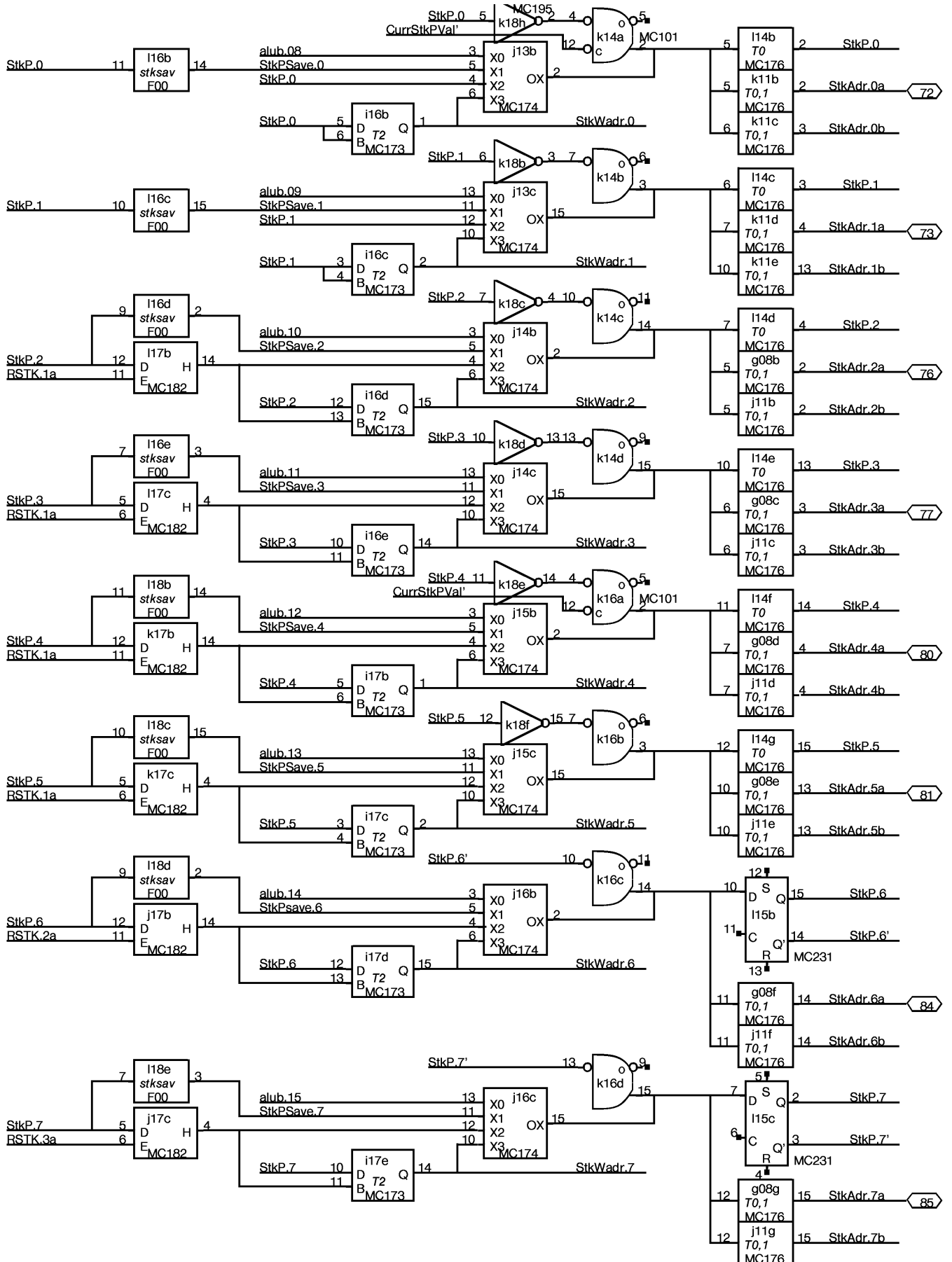


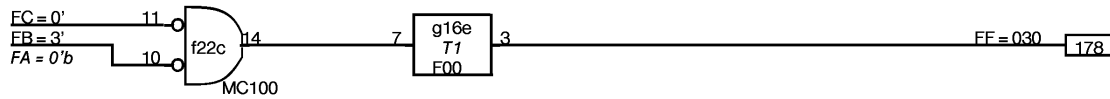


RBase bits that make up Rm address bits 0-3

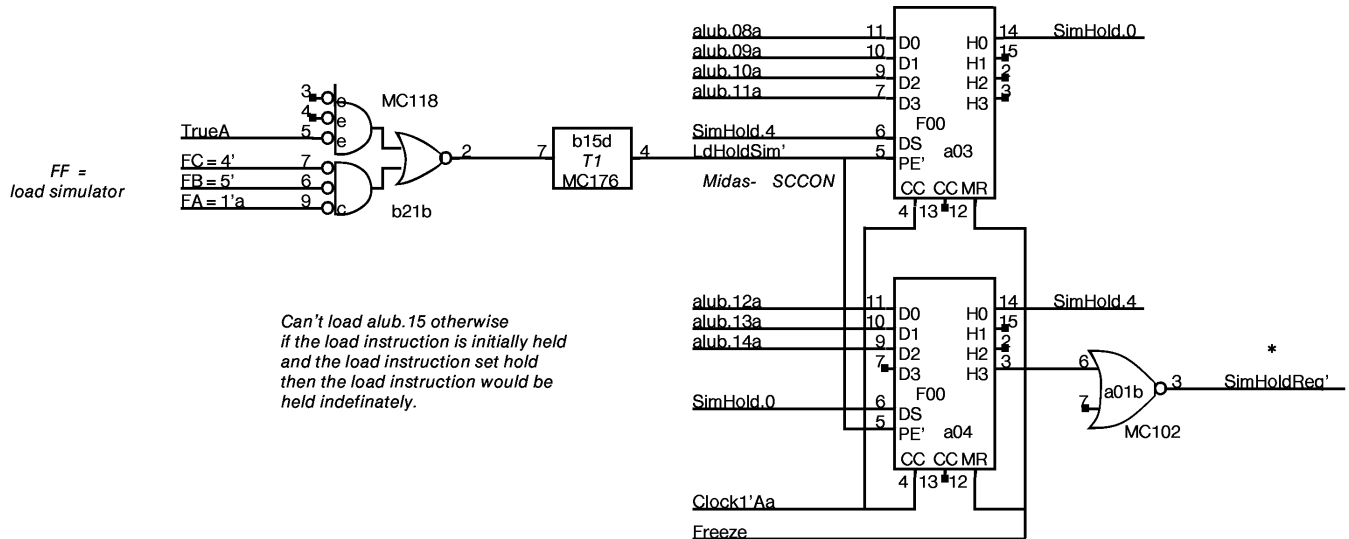
RSTK bits that make up Rm address bits 4-7







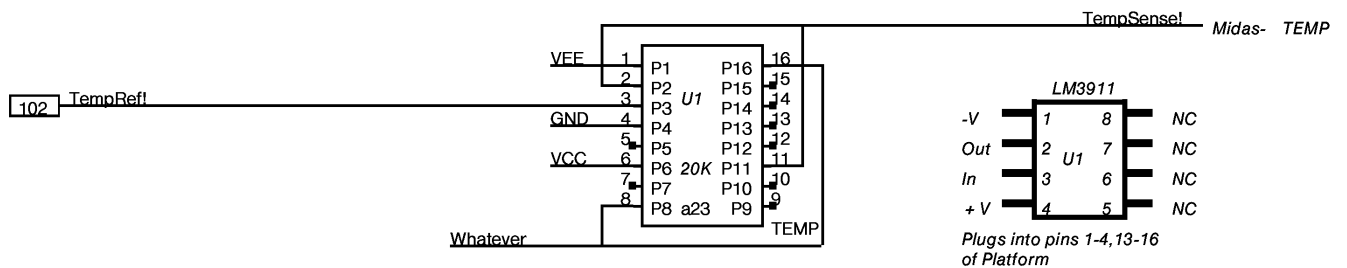
Spare FF Decode

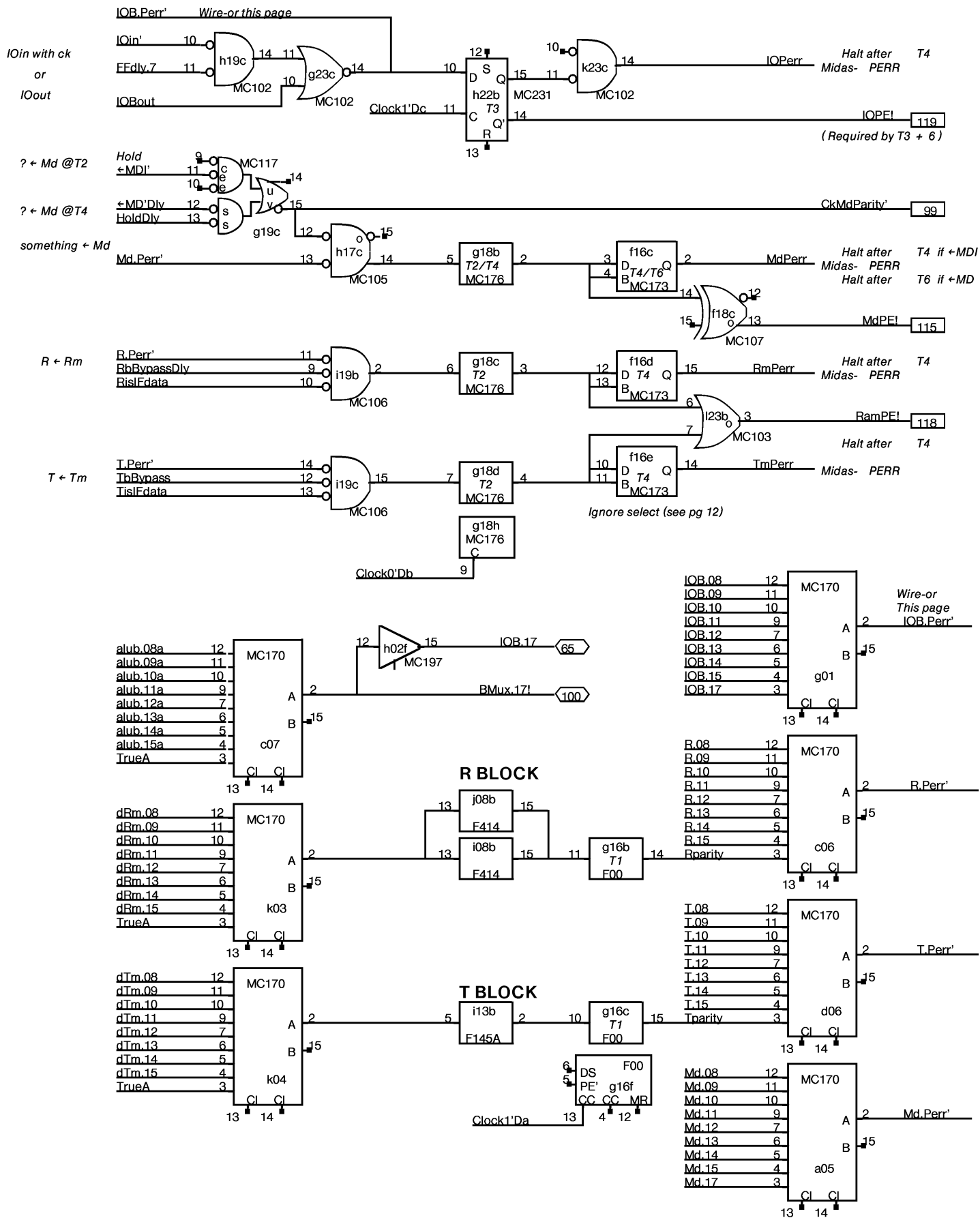


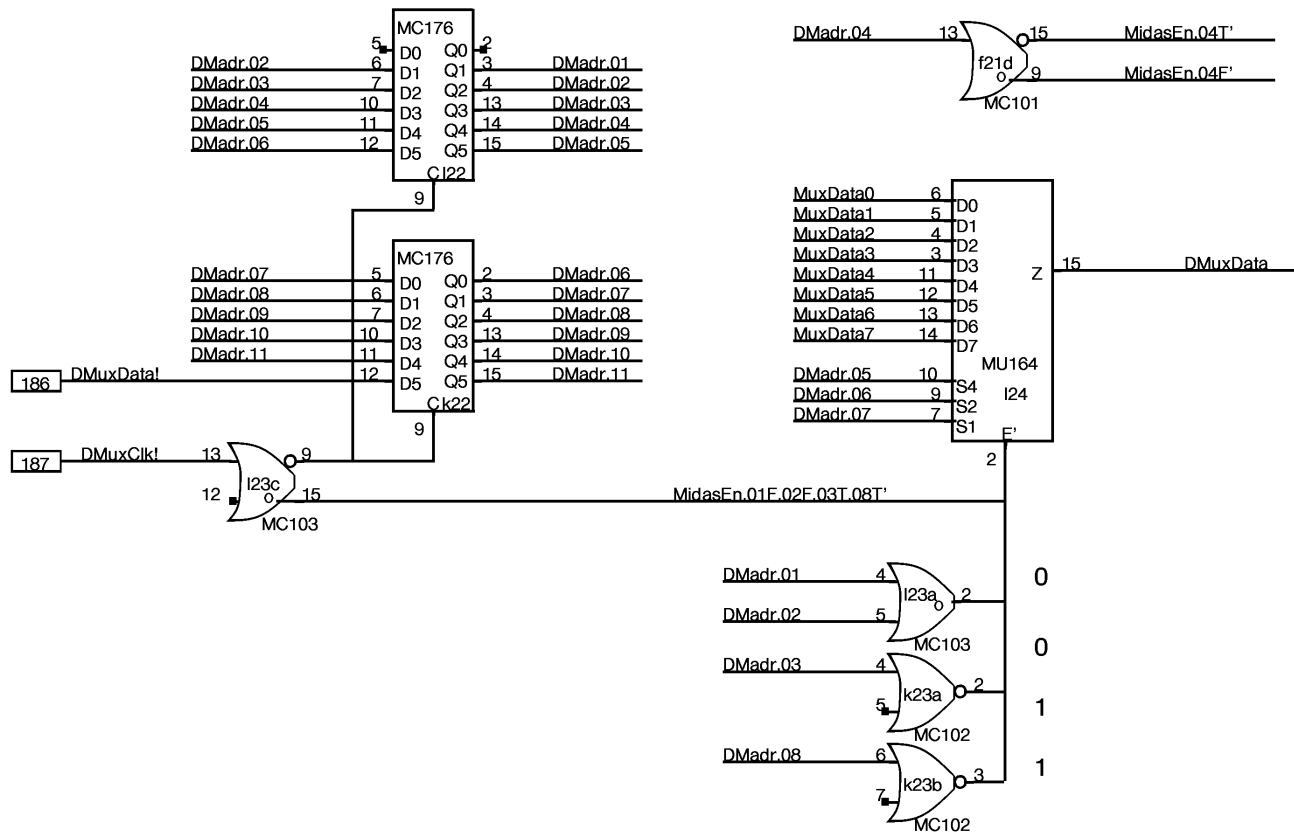
* NOTE: wire [086] with 71 ohms to Gnd
to disable the simulator. (see page 24)

Hold Simulator

Temperature Sensing Ckt



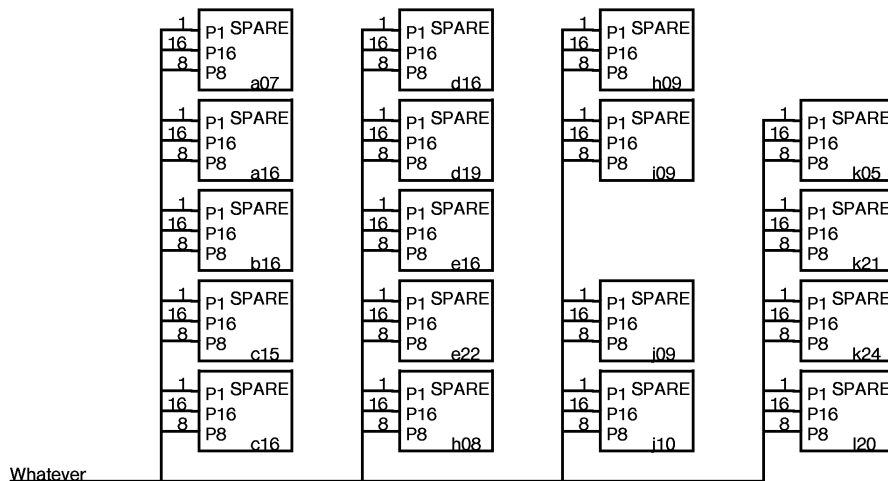




Board B in right half of ALTO word
Board A will be in the left half

Midas Control

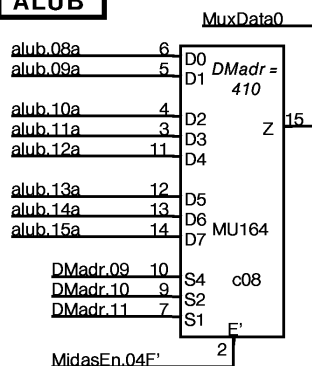
Spair Socket Declarations



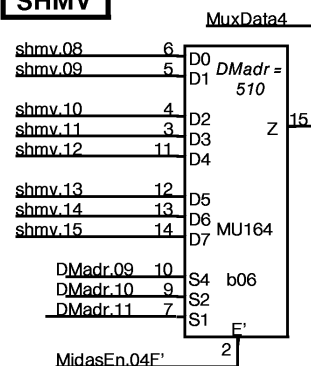
These should be the entire collection of unused locations.

This declaration will cause holes to be drilled in these locations on multiwire boards.

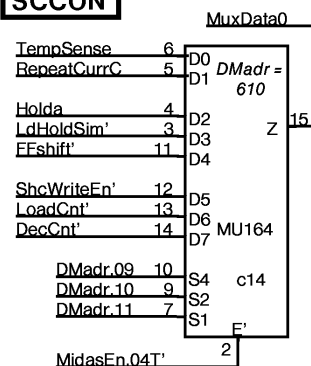
ALUB



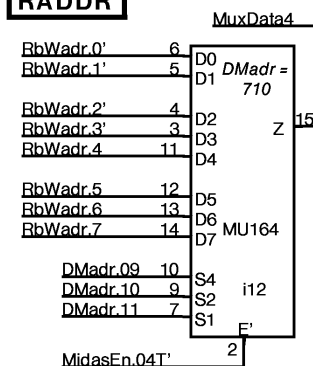
SHMV



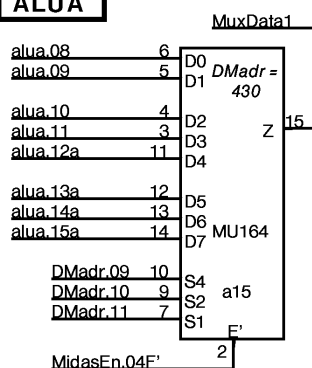
SCCON



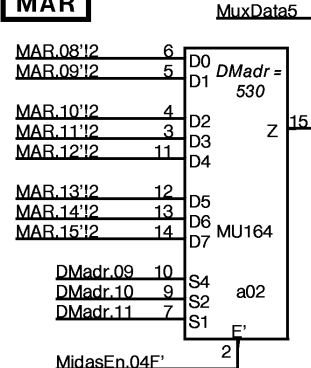
RADDR



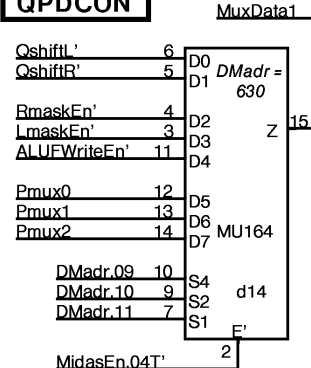
ALUA



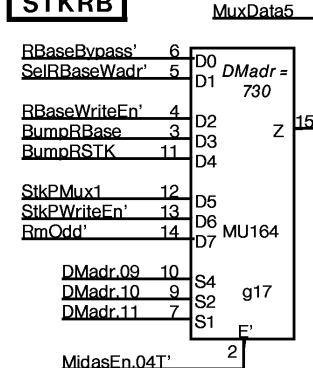
MAR



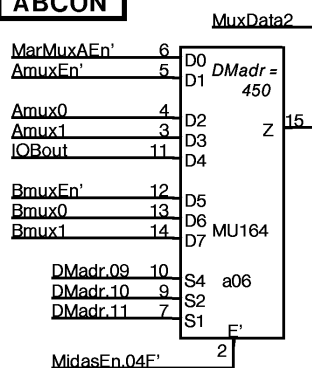
QPDCON



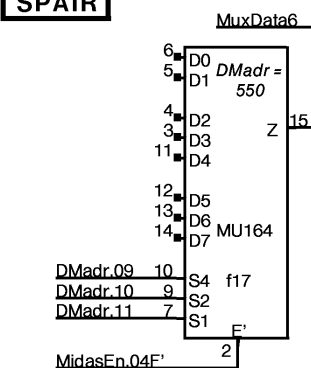
STKRB



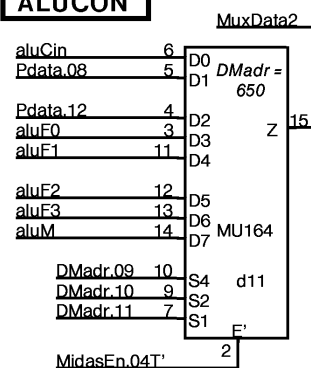
ABCON



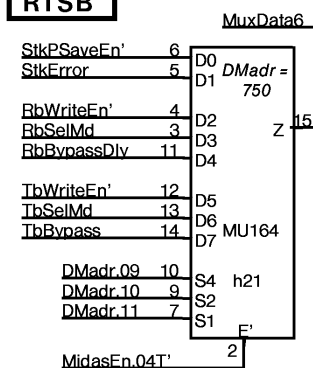
SPAIR



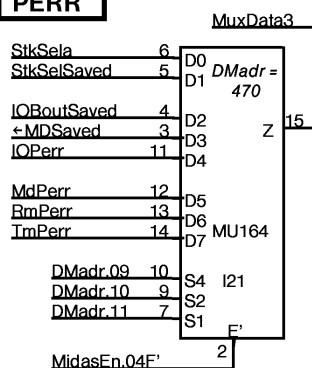
ALUCON



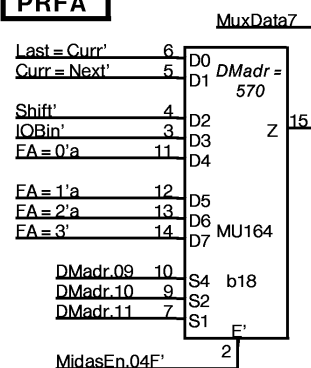
RTSB



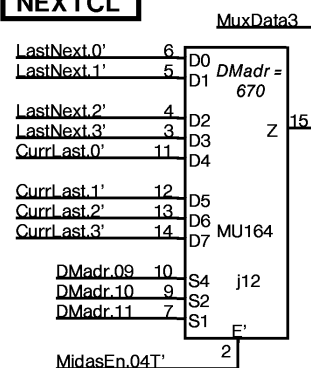
PERR



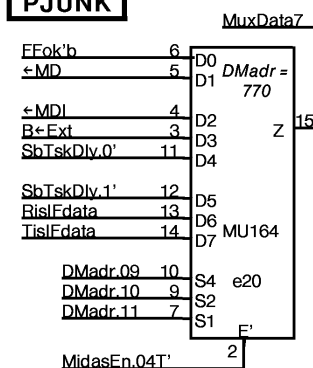
PRFA

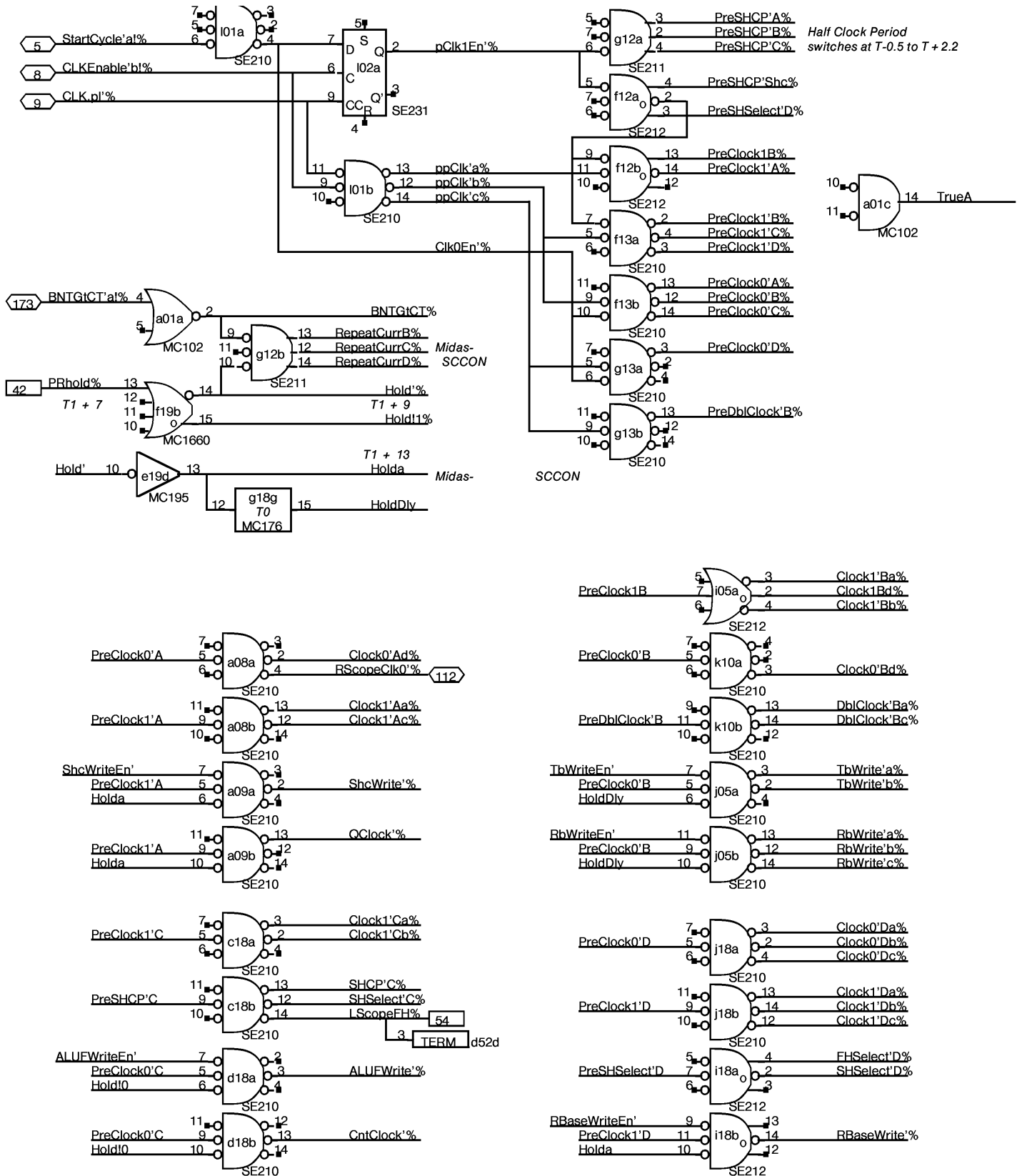


NEXTCL



PJUNK





-5v

+5v

Spare = 19

XEROX
PARC

Project
Dorado

Reference	Board Layout
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File
ProcL31.sil

Designer
R. Bates

Rev
Da

Date
1/21/80

Page 31

Use Dorado Proms to define the following Proms:

Board Name	Prom Name	location
PorcL	Lmask (low bite)	b07
	Rmask (low bite)	b08